

# Everyone Was Right: Lexical Representations in Language Models Are Both Abstract and Holistic

Anonymous ACL submission

## Abstract

Lexical representations are central to linguistic theory, yet to what extent they’re memorized, item-specific representations or abstract, generalizable representations remains a core debate. In the present study, we investigate how Large Language Models (LLMs) navigate this distinction using a quantitative framework that decomposes contextual word representations into three additive components: an item component, a class component, and an idiosyncratic word-class interaction. After validating our measurement on an artificial language, we evaluate six pretrained transformers (OPT-BabyLM and Pythia families) across two linguistic categories: part of speech and grammatical role. We find that the idiosyncratic interaction term is consistently positive and grows substantially across layers. Crucially, while abstract lexical and class axes remain nearly orthogonal, the interaction term systematically overlaps with both. Together, our results demonstrate that LLMs do not commit exclusively to pure abstraction or exemplar storage; rather, they construct representations that are simultaneously both abstract and holistic.

## 1 Introduction

Lexical representations are deceptively complex: a single word often carries vastly different meanings depending on the context, yet these meanings remain recognizably related. The categorization literature has long recognized this tension, noting that assigning a novel meaning to a specific category (or word) is rarely straightforward. Simple rule-based approaches, such as rigid feature lists, quickly break down: a three-legged cat is undeniably a cat, even if it violates one of the criteria used to define the category. Ultimately, defining a lexical category in precise terms may be nearly impossible. Yet, as U.S. Supreme Court Justice Potter Stewart famously observed in another context, we

may not be able to define a category precisely, but we nevertheless “know it when [we] see it.”<sup>1</sup>

This central paradox, that representations are simultaneously context-specific yet broadly generalizable, has fueled long-standing debates over the nature of lexical representations. The two leading frameworks offer distinct solutions: **abstraction theories** posit that humans construct generalized mental representations stripped of specific context, while **exemplar theories** argue that representations are built from distinct, context-rich memory traces.

In the present study, we investigate how Large Language Models (LLMs) navigate this paradox. Specifically, we examine the extent to which LLM lexical representations comprise abstract, generalizable features versus context-specific exemplars.

### 1.1 Evidence for Abstractions

Abstraction theories offer an intuitive account of the context invariance of language. At every level of language, there is substantial variability across contexts, from phonetic differences between speakers, to syntactic constructions that can take on a variety of lexical items, to the same word taking on different meanings in different contexts. The existence of abstractions accounts for the ability of humans to generalize across these contexts and recognize the same underlying representation despite this variability. Evidence for abstraction comes from linguistic productivity (Berko, 1958), semantic extension (Harmon and Kapatsinski, 2017), priming, word order preferences, and other phenomena.

Even something as deceptively simple as word order preferences, such as binomial ordering preferences, is shaped by abstract preferences (Morgan and Levy, 2015, 2016, 2024). For example, humans have strong preferences for binomial order-

<sup>1</sup>In *Jacobellis v. Ohio* (1964), Justice Stewart famously addressed the difficulty of defining obscenity, remarking that while formulating a precise definition might be a herculean task, he could still recognize obscene material when he saw it.

ings such as *bread and butter* over *butter and bread*, or *bishops and seamstresses* over *seamstresses and bishops*. Those preferences are to some extent driven by abstract preferences such as a preference for male-coded words before female-coded words. Crucially, speakers’ preferences for binomials that they have never encountered before are driven by these abstract properties, so the preferences cannot be stored with any particular pair of words.

## 1.2 Evidence for Exemplars

Exemplar theories, in contrast, argue that abstract representations are not necessary. Instead, linguistic knowledge may be stored as context-rich memory traces whose properties vary with experience. Evidence for this view comes from both the gradient perception of phonetic cues (McMurray et al., 2008) and the representation of larger linguistic units, including multi-word phrases (Bybee, 1999; Houghton and Morgan, 2023; Houghton et al., 2025, 2026; Morgan and Levy, 2016, 2024).

More specifically, there’s evidence that humans are sensitive to within-category changes in voice onset time (McMurray et al., 2008), produce *don’t* in a more reduced manner in high-frequency phrases than in lower-frequency phrases (Bybee, 1999), have stronger word order preferences for high-frequency binomials (Morgan and Levy, 2024), and recognize *up* less well in high-frequency and high-predictability V+*up* phrases (Houghton, 2025; Kapatsinski, 2010). These findings suggest that linguistic representations retain information specific to the contexts in which their components occur, rather than being fully abstracted across contexts.

Most of the evidence above concerns sequences of words, but the same question arises within a single word. Whether regularly inflected forms are retrieved whole or assembled from a stem and an affix has been debated on much the same grounds, with frequency again playing a central role (Stemberger and MacWhinney, 2004, 1986). But the same logic applies even when there is no overt morphological combination. *Walk*, for example, occurs as both a noun and a verb. If the representation of *walk* as a verb was fully determined by a representation of *walk* together with a representation of the verb category, then nothing in that representation would be specific to that pairing. If instead some part of the representation is unique to *walk* as a verb, then the representation contains information about a particular combination that cannot be

reduced to its parts.

## 1.3 Present study

The evidence for both abstraction and item-specificity therefore raises a broader question about the structure of linguistic representations: is a representation of a specific linguistic construction a combination of abstract features, or does that representation also contain item-specific knowledge that is irreducible to these component parts? This is difficult to settle behaviorally, since a learner with no abstract representations at all can look like one that abstracts (Houghton and Kapatsinski, 2026), but it is a property of a representation, and unlike in humans, model representations can be examined directly (Elhage et al., 2022; Pluth et al., 2026).

We hypothesize that linguistic representations can contain both abstract and item-specific information. We therefore decompose a word-in-class representation into three components: the item-component, the component that generalizes across different classes of the same word (the part of *walk* as a noun that is shared by *walk* as a verb), the class-component, the component that generalizes across all items of the same class (e.g., the part of *walk* as a noun that is shared by *run* as a noun, and *plant* as a noun) and what belongs to that word in that class alone (e.g., the part of the representation unique to *walk* as a noun). We first validate this decomposition on an artificial language with controlled representational structure, then apply it to six pretrained language models across part of speech and grammatical role.

Our main contributions are as follows:

- We introduce a method for decomposing word-in-context representations into lexical, grammatical, and word-class components, and validate it on an artificial language.
- Across six pretrained models, the word-class component is consistently present and grows with depth; in the artificial language its size is governed more by representational capacity than by the language’s composition. The lexical and grammatical components occupy nearly separate axes, while the word-class component overlaps with both.
- We show that the word-class component contains information about the class itself beyond the shared class representation.

## 2 Experiment 1: Artificial Languages

In Experiment 1 we test our measurement on an artificial language whose features we control. This allows us to validate our measurement while simultaneously examining how different features of the language affect a model’s representations.

### 2.1 Dataset

In order to examine the separability of linguistic representations, we constructed an artificial language which comprises a probability distribution over a vocabulary of 600 tokens. Every observation is a combination of a **form** (one of 60), a **class** (one of 4), and a **context** (drawn from a pool of 400). The motivation for this setup is to give each word a naturalistic environment: the word occurs in a context that favors a specific class-membership (e.g., *walk* as a verb occurs in a different context than *walk* as a noun). Each form is observed 24 times in each class, each time in a different context, mirroring the patterns of human language. Consider *walk* used as a verb in one of those contexts. Four things determine the distribution over the next token. *Walk* itself contributes the same amount regardless of the class it is used in, and the verb class also contributes the same amount for every item. The third contribution is what is specific to *walk-as-a-verb* over and above those two: every form and every class has its own random vector, and this term multiplies *walk*’s by the verb’s, entry by entry. The context contributes the rest, which is noise as far as our question goes, though the model still has to encode it in order to accurately learn the language. The four are weighted, summed and passed through a softmax layer in order to create the ground truth probability distribution over next tokens for a given input:

$$P(\text{next token} \mid f, c, x) = \text{softmax} \left( s \left[ w_\alpha \text{item}(f) + w_\beta \text{class}(c) + w_\gamma \text{int}(f, c) + w_{\text{ctx}} \text{context}(x) \right] \right) \quad (1)$$

Each term is isotropically scaled to unit variance before it is weighted,<sup>2</sup> so changing the weights

<sup>2</sup>Specifically, each observation consists of a form in a specific class and a context, paired with that term’s contribution to the next-token distribution for that observation. We normalize this dataset by dividing each value in it by the total standard deviation across all observations. This is necessary because different matrices can contain the same information at

changes the mix of the three effects without changing how much signal there is in total.

The weights  $w_\alpha$ ,  $w_\beta$ , and  $w_\gamma$  in Equation 1 control how much of the true distribution each of the three effects contributes, and we cross them factorially over  $\{0, 0.5, 1, 2\}$ : zero removes an effect from the language altogether, and the other three values make an effect half as strong as another, equal to it, or twice it. Similarly,  $w_{\text{ctx}}$  governs how much the context contributes, so we cross those languages with three context weights:  $\{0.5, 2.0, 6.0\}$ , ranging from context being a minor part of the true distribution to the dominant one. The four weights are subsequently rescaled together to unit length, such that  $w_\alpha^2 + w_\beta^2 + w_\gamma^2 + w_{\text{ctx}}^2 = 1$ .<sup>3</sup> The ranks of the item and class are held fixed at 8 and 4 respectively but each language is crossed with an interaction effect that is chosen a set of four different ranks that it can take:  $\{4, 16, 64, 177\}$ . Additionally, we train a model on each of the resulting languages at hidden widths of 8, 16, 32, 128, 256, with either a linear or a ReLU activation. The whole grid is then run with five random seeds giving 37,800 runs in all, of which 99.6% trained to convergence; Appendix A lists every factor and its levels.

### 2.2 Method

All code and analyses in this paper can be found here: <https://github.com/znhoughton/representation-separability>.

#### 2.2.1 The learner

The learner model is a multi-layer perceptron (MLP) with a single hidden layer of width  $d$  and either a linear or a ReLU activation function. The model takes as its input three pieces of information: form, class, and context. Each has its own embedding, and the three are concatenated and passed through the model which then outputs a probability distribution over the vocabulary (Figure 1). The training objective is thus to minimize cross-entropy between the model’s predicted distribution over the vocabulary and the ground-truth distribution.

different scales. For example, imagine a matrix with random numbers, and that same matrix with each number multiplied by two. The information is the same, but the weights will have a larger effect on the latter dataset.

<sup>3</sup>This makes the resulting languages more comparable by enforcing the same total signal (i.e., the variance of the weighted sum in Equation 1 is held constant), despite individual differences in the amount that each individual term contributes.

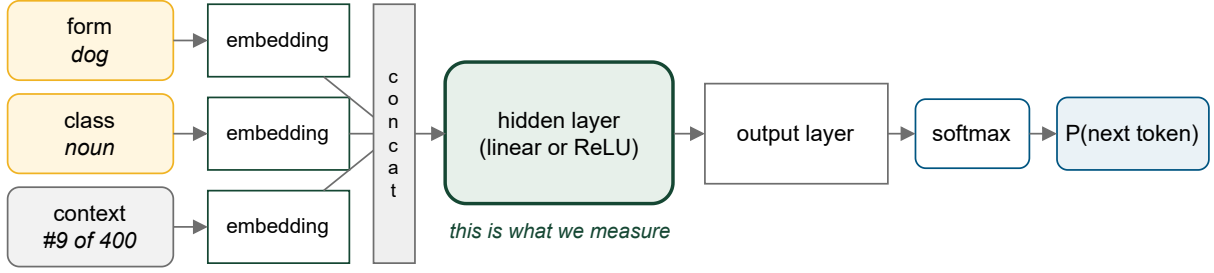


Figure 1: Structure of our model. Every form is observed 24 times at each class, each time in a different context; that variation across contexts is what the measurement later splits in two.

### 2.2.2 Decomposing a representation

We measure separability by decomposing a word-in-context representation into a lexical component, a class component, and an idiosyncratic component. The representation is the MLP’s hidden layer, which for a given form and class varies across the contexts it is observed in. Let  $M[i, c] \in \mathbb{R}^d$  be the mean of those hidden states for item  $i$  used at class  $c$ . We decompose those item-by-class means by centering, so that:

$$M[i, c] = \mu + \alpha(i) + \beta(c) + \gamma(i, c), \quad (2)$$

where  $\mu$  is the grand mean;  $\alpha(i)$  is item  $i$ ’s mean across classes minus  $\mu$  which conceptually is the representation unique to the item;  $\beta(c)$  is the mean across items at class  $c$  minus  $\mu$  which conceptually is the representation unique to the class (e.g., noun vs verb); and  $\gamma(i, c)$  is what item  $i$  at class  $c$  holds that neither marginal accounts for which conceptually is the item-specific representation in a specific class (Figure 2). Since every item is observed in every class with equal weight, averaging over class isolates the item effect, and averaging over items isolates the class effect. Each representation for a given item in a given class is exactly the sum of the four terms and no two of them vary together across those representations, so the squared lengths of each term (excluding  $\mu$ ) sum to the total and we report each as a share.

Each of these averages is taken over a sample of observations that differ only in context, so each carries some of the context term of Equation 1 as noise. We remove it by dividing the observations into two evenly sized splits, running the decomposition on each split separately, and estimating each component’s size as the dot product of its two split estimates, which is large when the splits agree about the component and near zero when they do

not. As a result, noise that differs between the splits is independent and cancels.

For each representation we report two measurements. The first measure is the fraction of the representation accounted for by lexical, class, and idiosyncratic information: each component’s squared length, summed over its vectors and divided by the total, so that the three fractions sum to one. The second measure quantifies the extent to which those components occupy the same directions. Specifically, we project the item vectors onto the class direction and record the fraction of their squared length that survives, and we project the interaction onto the smallest subspace containing both the item and the class vectors (i.e., the span of the two subspaces).<sup>4</sup> A representation is separable if a word’s representation is the sum of an item part and a class part, each on its own axis. Separability can fail in two ways: the parts may be non-orthogonal, so that projecting out the class also removes item information, or there may be a third component that is neither, regardless of its orientation. The question is accordingly not simply whether class information is present, but how it is represented: on a single direction shared across words, or on a separate direction for each of them.

To quantify whether a size is meaningful, we compare each against zero, which is the null because the dot product of two independent split estimates has an expected value of zero if there is no structure in the underlying representations. We redraw the split two hundred times and call a size meaningful when its confidence interval excludes zero; for components whose value is exactly zero this flags an effect about 5% of the time (Appendix

<sup>4</sup>The class effect spans at most one direction fewer than the number of classes, so it is low-dimensional in both experiments: one direction in Experiment 2 and at most three in Experiment 1. Its projection into the item subspace, which has many more dimensions, has high overlap by chance; we therefore do not report it.

B), the false-positive rate expected of a 95% confidence interval.

To quantify whether an overlap is meaningful, we project the effect, held exactly as measured, onto two hundred random subspaces of the same rank, which gives the distribution of overlaps an unrelated subspace would yield. An overlap below that baseline indicates meaningful orthogonality; an overlap above the baseline indicates meaningful non-orthogonality. We report each overlap as a multiple of the baseline’s median, with how many measurements fall outside its 95% interval; Appendix C gives the algebra for both measures.

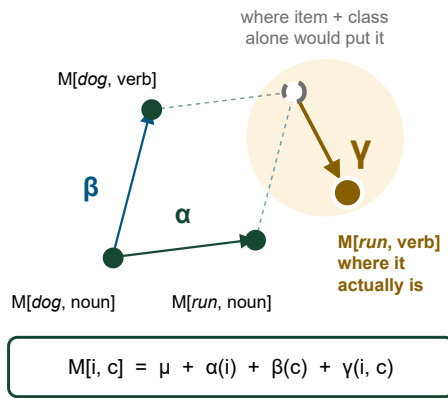


Figure 2: The four item-by-class means formed by two items in two classes.

It’s important to note that the weights in Equation 1 describe the distribution the model is trained to match, while the measure decomposes the representation that the model learns, so the two sets of numbers are not on a common scale: a learner handed a language that is 85% interaction can represent it with the interaction taking a far smaller share. Appendix B reports a separate check in which components were set directly in a representation and recovered almost exactly, which is what licenses interpreting the results below as facts about the learner rather than about the measure. We include runs only if at least one of the three components has a re-split interval excluding zero; where none does, the three shares divide noise.

### 2.2.3 Analysis

We model our data using an ordered beta regression (Kubinec, 2023) in *brms* (Bürkner, 2017) with default priors, since our fractions lie between 0 and 1 *inclusively* (which a simple beta regression can’t model). We modeled each component’s fraction of the representation as a function of its own effect

size in the language, the hidden width, and the total rank of the planted structure (with interactions between each), with the other two effect sizes as controls, and random intercepts by language. Only the interaction’s rank varies, so that total moves with it alone. Equation 3 gives the item model; the class and interaction models are analogous, but with their own effect being included as an interaction term. All predictors are centered, and a language a specific combination of parameters, so its five seeds share an intercept.

$$\begin{aligned} \text{ItemFraction} &\sim \text{ItemEffect} * \log_2(\text{Width}) \\ &* \log_2(\text{TotalRank}) \\ &+ \text{ClassEffect} + \text{InteractionEffect} \\ &+ (1 \mid \text{Language}) \end{aligned} \quad (3)$$

For our second measure, we modeled each overlap as a logged multiple of its own chance level with a Gaussian family. An overlap is a property of two components jointly, so there is no own effect size to cross with the architecture: all three enter as controls and only width and rank are crossed (Equation 4). All predictors are centered for this model as well.

$$\begin{aligned} \log(\text{Overlap}/\text{Chance}) &\sim \text{ItemEffect} \\ &+ \text{ClassEffect} + \text{InteractionEffect} \\ &+ \log_2(\text{Width}) * \log_2(\text{TotalRank}) \\ &+ (1 \mid \text{Language}) \end{aligned} \quad (4)$$

## 2.3 Results

Results are shown in Figure 3, with every coefficient in Appendix D.1 and the remaining properties of the runs in Appendix D.

Each effect’s size in the language raises its own size in the representation and lowers the other two, as three shares of a single total must, so the decomposition is specific to the component it names, and of the three the item effect takes the largest share at every width. This holds under both activations: even the linear learner, which has no parameter depending on the form and the class together beyond a linear combination of them, builds more of the interaction the more the language contains ( $\beta = +2.070$ ).

The extent to which the interaction is encoded in the model’s representation does not depend on its rank in the language ( $\beta = -0.016$ ). Instead, it is the size of the hidden layer that determines how

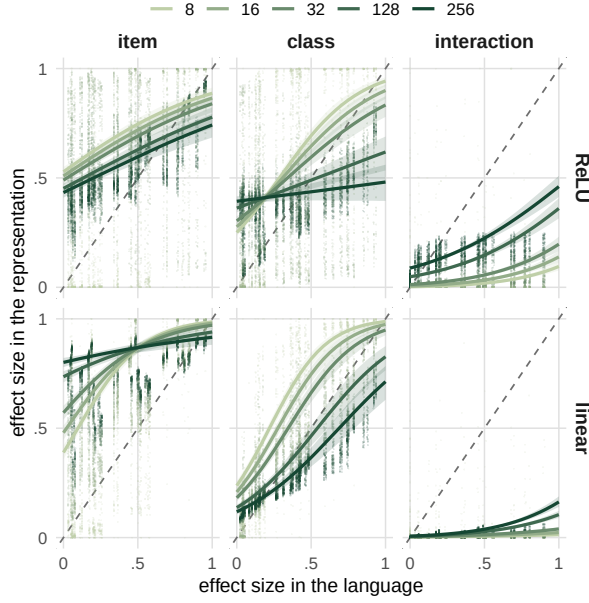


Figure 3: Each component’s share of the representation against its share of the language, by activation. Faint points are runs; lines are fitted values from Equation 3 with 95% intervals, coloured by hidden width. The dashed diagonal is exact reproduction of the language’s composition.

much of the interaction is encoded ( $\beta = +0.568$ ): the larger the hidden layer, the larger the size of the interaction it encodes. This is not the case for the item effect, which decreases as the hidden layer grows ( $\beta = -0.095$ ), or for the class effect, which barely changes ( $\beta = +0.030$ ). In the linear condition the hidden layer is spent the other way, on the item effect, leaving the interaction near zero throughout; rank does register there ( $\beta = -0.084$ ), but well below width ( $\beta = +0.187$ ). Under a ReLU learner the interaction does not fall away even where the language is additive, sitting at 0.012 against 0.004 for the linear learner (the left edge of Figure 3’s interaction panel; Appendix D gives the rates at which each is detected at all). The model only ever encounters words in specific contexts, so the additive structure of the language is visible across the corpus rather than in any single observation.

Width governs geometry as well as magnitude. The width coefficient on item-class overlap is large and negative and excludes zero ( $\beta = -0.398$ ): the wider the layer, the less the item and class effects share directions, falling from above chance to below it. The interaction’s overlap with the two margins carries a coefficient of the same sign and nearly the same size ( $\beta = -0.315$ ), so both geometries open up together as the layer grows. Under

a linear learner the two come apart. The item-class coefficient is small, negative, and includes zero ( $\beta = -0.002$ ), so width leaves that overlap where it is; the interaction’s overlap with the margins instead carries a small positive coefficient that excludes zero ( $\beta = +0.101$ ), opening up as the layer grows. Neither rises above chance at any width under the linear learner (Figure 9). A narrow layer thus does not merely represent less of the interaction, it also forces the components it does represent onto shared directions, and given room the learner separates them by more than chance. Every coefficient of these models, with its interval, is in Appendix D.2.

Which components are forced together depends on which subspaces compete, not on the total structure. The item and class effects need a fixed twelve directions between them however much interaction structure the language holds, and their overlap accordingly crosses chance at nearly the same width whether the interaction spans 4 dimensions or 177. A large interaction runs out of room for itself well before it pushes the item and class effects together. How much of each component the model builds, and how separately it keeps them, therefore depends on how wide the hidden layer is rather than on how many dimensions the interaction spans.

### 3 Experiment 2: Language Models

Experiment 1 established that if the language a learner model encounters is not purely additive, the model will represent the language in a non-separable manner, with the extent governed by the width of the model. Experiment 2 applies the same decomposition to the contextual representations of pretrained language models, across two linguistic categories: part of speech and grammatical role.

#### 3.1 Dataset

In order to examine the separability of linguistic representations in models, we examined two linguistic categories: part of speech and grammatical role. For both, we concatenate six general-domain English Universal Dependencies treebanks (Nivre et al., 2020): EWT (Silveira et al., 2014), GUM (Zeldes, 2017), LinES (Ahrenberg, 2007), ParTUT (Sanguinetti and Bosco, 2015), GENTLE (Aoyama et al., 2023), and PUD (Zeman et al., 2017), giving 41,095 sentences and 706,323 tokens. We measure two model families at three matched scales: OPT-BabyLM, trained on devel-

omponentally plausible quantities of child-available language (Houghton et al., 2026; Warstadt et al., 2023), and Pythia, trained on the Pile (Biderman et al., 2023), at 125M/160M, 350M/410M, and 1.3B/1.4B parameters, with widths of 768, 1024, and 2048. Appendix E gives parameter counts and training-token estimates.

### 3.2 Stimuli

Our stimuli comprise two linguistic categories: part of speech (NOUN versus VERB) and grammatical role (*nsubj* versus *obj*, restricted to nouns). For each category we select items that occur in both of its classes, such as *walk*, which occurs as both a noun and a verb. An item is a single inflected word form, not a lemma, so different inflections of one lemma are separate items: *walk* and *walking* are both items, each included only if it is itself attested in both classes. Pooling them under one lemma would collect *walk*, *walks*, *walked*, and *walking* into a single item, and the two classes would then be read at different tokens, which the embedding layer could then separate before any contextualization has occurred. Capitalization is ignored when occurrences are grouped, but not in the text the models read, so a sentence-initial *Walk* counts toward the item *walk* and is tokenized as it appears there.<sup>5</sup>

An item is included only if it occurs at least ten times in each class; forms occurring in only one are excluded, which bites harder for part of speech, since fewer forms occur as both noun and verb than nouns occur as both subject and object. This yields 78 items for part of speech and 49 for grammatical role, with median counts per class running from 15 to 19 (Table 14), inside the range we validated for the measurement (Appendix B).

### 3.3 Method

The decomposition, the two reported quantities, and the replication logic are identical to Experiment 1. For every token we take the hidden state of the word’s **last subword** at every layer of each model (using the model’s own tokenizer).<sup>6</sup>

<sup>5</sup>OPT-BabyLM’s tokenizer is uncased, so this choice has no effect for that family of models. On the other hand, Pythia’s tokenizer is not uncased; however, lowercasing the input would be the larger distortion, since a sentence-initial word is capitalized in written English which the model was trained on.

<sup>6</sup>Almost every item is a single subword, so this choice has little bearing on the results: zero items are split into multiple subword tokens under Pythia’s tokenizer, and only eight are split into multiple subword tokens under OPT-BabyLM’s tokenizer.

Transformer hidden states contain a few *rogue* dimensions whose variance exceeds the rest by two to three orders of magnitude (Timkey and van Schijndel, 2021), and these would otherwise dominate any subspace estimated from the raw scatter, so we divide each dimension by its standard deviation before decomposing (Appendix G).

### 3.4 Results

The results for the final layer are reported in Table 16; the layer-by-layer results are shown in Figure 4.

The item effect is the largest component in both categories, running from 0.657 to 0.846 for part of speech and 0.828 to 0.904 for grammatical role. What differs between them is how the remainder divides. For grammatical role the class effect, at 0.039 to 0.078, is comparable to the interaction at 0.045 to 0.094; for part of speech the interaction, at 0.096 to 0.245, runs well above the class effect at 0.050 to 0.098.

These are shares of the item-by-class structure, which say how that structure divides rather than how much of a token’s representation it accounts for. Placing the three on one scale, the reduction in squared error when predicting a token’s hidden state against guessing the grand mean, knowing which word a token is accounts for 19% to 38% of that error, knowing which class it is used in a further 1% to 3%, and knowing the word-class combination another 2% to 8% on top of both. The item-specific component therefore contributes at least as much as the class it sits inside in 11 of the twelve grids, and roughly three times as much for part of speech. Appendix H gives the figures per model.

Both the class effect and the interaction grow stronger through the layers of the model. The interaction is essentially absent at layer 0 in both categories, between 0.0009 to 0.0088 across every category and model, and rises to between 0.0455 to 0.2449 (Figure 4, top row). This is not surprising since at the first layer, the model has yet to integrate enough context into the representation to commit to the class of the representation. What the network builds over those layers is also not an artifact of where a word sits in the sentence: zeroing OPT-BabyLM’s learned absolute position embeddings leaves the class structure more word-specific rather than less, and Pythia, which has no such embeddings to remove, shows the interaction at least as strongly (Appendix I).

Turning to how those components are oriented, the class direction carries almost none of the item effect. At the deepest layer, item-class overlap runs 0.0035 to 0.0067 for part of speech and 0.0033 to 0.0062 for grammatical role, so projecting out the class direction leaves more than 99% of the item effect intact. While both overlaps are technically less orthogonal than a random subspace, the overlap itself is tiny, so the two are near-orthogonal rather than exactly orthogonal.

In contrast to the main effects, the interaction is not orthogonal to class or item. Across the six models, 0.106 to 0.239 of the interaction for part of speech is in the span of the two main effects, and for grammatical role 0.071 to 0.179 is, against nulls whose upper ends run 0.023 to 0.058 and 0.014 to 0.038 respectively. The smallest of these overlaps exceeds the largest of the nulls, so the pattern holds in every model. In other words, the small departure from orthogonality between item and class becomes a large one for the interaction.

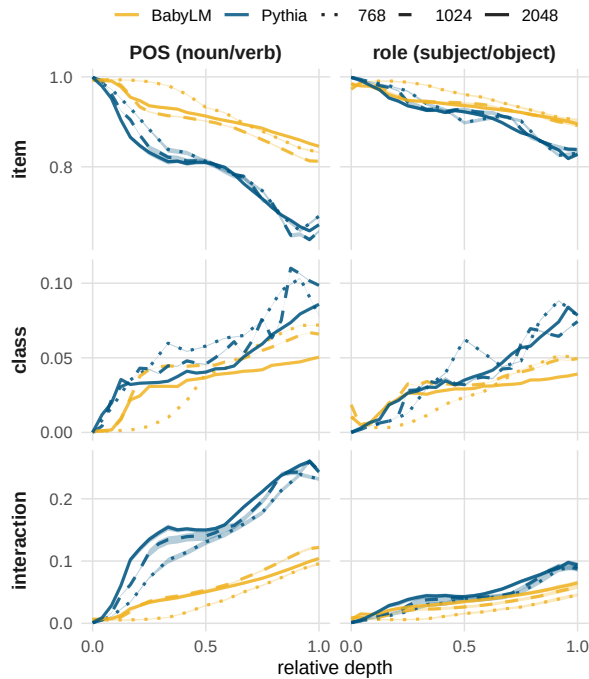


Figure 4: The three components through the network, by linguistic category. x-axis is layer depth. The y-axis is the effect size for item, class, and the interaction. Each line is one model: color is the training corpus, linetype the hidden width; the band is that model’s re-split 95% interval.

## 4 Experiment 3: Probing the Interaction

Experiment 2 shows that the interaction is large and that it replicates across independent splits of

a word’s occurrences, but neither shows what information is contained within the interaction. An item-by-class term captures any reliable difference between a word’s uses in the two classes, and some of that difference may be correlated with the context, since subjects appear in different constructions from objects.

### 4.1 Method

We test whether the interaction encodes class information with a probe. If  $\gamma$  encodes the class, the class should be recoverable from the part of a token’s representation that is neither the item effect nor the shared class direction, and we remove both explicitly. All probing happens *within* a single word, so we ask whether *walk*’s noun uses can be told from *walk*’s verb uses; the item effect is constant within a word and so cannot contribute. We then estimate  $\beta$  as the difference between the two class means pooled over all items, normalize it, and remove that component from every token, so that whatever separates the classes afterwards is not the direction they share.

What remains is the interaction: the part of a word of a specific class that is neither a fact about the word nor a fact about the class. We fit a cross-validated logistic probe to it, five folds, with classes balanced by subsampling within each word so that chance is 0.50, and average the resulting accuracies over words, with a word contributing only if it has at least ten tokens in each class, the same threshold the stimuli are selected on. Some words rest on few tokens, so the mean carries a percentile bootstrap interval over words. The whole procedure is then repeated with the class labels permuted within each word, which preserves every word’s token count and class balance while destroying the labeling, and that null is where a probe reading its own cross-validation folds would land (Hewitt and Liang, 2019). We run this at the deepest layer of the largest model in each family, for both categories.

### 4.2 Results

Results are reported in Table 1.

For both families of LLMs across both linguistic categories, the class is recoverable from the interaction at well above chance while the permuted null sits on chance, reaching 0.94 for part of speech and 0.84 for grammatical role, with bootstrap intervals excluding the nulls in all four cases. The null landing on chance is what makes those accuracies interpretable, since it is the same probe, the same

| class            | model       | accuracy          | null              |
|------------------|-------------|-------------------|-------------------|
| part of speech   | BabyLM 1.3B | 0.90 [0.88, 0.91] | 0.51 [0.49, 0.54] |
| part of speech   | Pythia 1.4B | 0.94 [0.92, 0.95] | 0.50 [0.48, 0.53] |
| grammatical role | BabyLM 1.3B | 0.80 [0.77, 0.83] | 0.50 [0.47, 0.53] |
| grammatical role | Pythia 1.4B | 0.84 [0.81, 0.86] | 0.52 [0.49, 0.55] |

Table 1: Accuracy of recovering the class from the interaction, at the deepest layer of the largest model in each family.

folds, and the same word-level averaging with only the labeling destroyed.

The probe accuracies follow the same ordering across categories as the interaction sizes (Table 16), providing converging evidence from the geometric and functional measures.

The results demonstrate that  $\gamma$  carries class information rather than generic correlated context, since generic context would not survive holding the word fixed and removing the shared axis. It does not show that  $\gamma$  is *only* the label: context correlated with the class within a word survives both removals, so the interaction remains an upper bound.

## 5 Discussion and Conclusion

The present study examined whether a word’s representation in a language model is exemplar-like or abstract and found that it contains elements of both. We decomposed a word’s representation in a class into an item effect, a class effect, and a word-class interaction. The first two capture information shared across classes or words, respectively, while the interaction captures information specific to a particular word-class combination. Across six pretrained language models and two linguistic categories, this interaction was present in every model and category, nearly absent at the embedding layer, and grew throughout the network. The abstract components were nearly orthogonal, whereas the interaction overlapped with both, indicating that abstract and item-specific information are represented together rather than on separate axes.

Overall, the central result is not simply that language-model representations are sensitive to context, but that this context sensitivity is organized around particular word-class combinations. Language-model representations therefore combine abstraction with item-specific structure: they generalize across words and classes while retaining information about particular combinations.

## Limitations

Our evaluation is restricted to English and relies on two linguistic categories tested across two model families (OPT-BabyLM and Pythia) at three parameter scales. While these choices cover a range of architectures and linguistic phenomena, broader typological, categorical, and architectural validation remains a direction for future work.

## References

- Lars Ahrenberg. 2007. [LinES: An English-Swedish parallel treebank](#). In *Proceedings of the 16th Nordic Conference of Computational Linguistics (NODAL-IDA 2007)*, pages 270–273, Tartu, Estonia. University of Tartu, Estonia.
- Tatsuya Aoyama, Shabnam Behzad, Luke Gessler, Lauren Levine, Jessica Lin, Yang Janet Liu, Siyao Peng, Yilun Zhu, and Amir Zeldes. 2023. [GENTLE: A genre-diverse multilayer challenge set for English NLP and linguistic evaluation](#). In *Proceedings of the 17th Linguistic Annotation Workshop (LAW-XVII)*, pages 166–178, Toronto, Canada. Association for Computational Linguistics.
- Jean Berko. 1958. [The child’s learning of english morphology](#). *WORD*, 14(2-3):150–177.
- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. 2023. [Pythia: A suite for analyzing large language models across training and scaling](#). In *Proceedings of the 40th International Conference on Machine Learning (ICML)*.
- Paul-Christian Bürkner. 2017. [brms: An R package for Bayesian multilevel models using Stan](#). *Journal of Statistical Software*, 80(1):1–28.
- Joan Bybee. 1999. [Usage-based phonology](#). *Functionalism and formalism in linguistics*, 1:211–242.
- Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, and 1 others. 2022. [Toy models of superposition](#). *Transformer Circuits Thread*.
- Zara Harmon and Vsevolod Kapatsinski. 2017. [Putting old tools to novel uses: The role of form accessibility in semantic extension](#). *Cognitive Psychology*, 98:22–44.
- John Hewitt and Percy Liang. 2019. [Designing and interpreting probes with control tasks](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2733–2743, Hong Kong, China. Association for Computational Linguistics.

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 740 | Zachary Nicholas Houghton. 2025. <i>Multi-Word Representations in Minds and Models: Investigating the Storage of Multi-Word Phrases in Humans and Large Language Models</i> . Ph.D. thesis.                                                                                                                                                                                                                                                                                                                                                                                                                        | 796 |
| 741 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 797 |
| 742 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 798 |
| 743 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 799 |
| 744 | Zachary Nicholas Houghton and Vsevolod Kapatsinski. 2026. Exemplars in disguise: Pure exemplar models mimic abstraction-first learning. <i>Preprint</i> , arXiv:2608.00821.                                                                                                                                                                                                                                                                                                                                                                                                                                        | 800 |
| 745 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 801 |
| 746 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 802 |
| 747 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 803 |
| 748 | Zachary Nicholas Houghton and Emily Morgan. 2023. Does predictability drive the holistic storage of compound nouns? In <i>Proceedings of the Annual Meeting of the Cognitive Science Society</i> , volume 45.                                                                                                                                                                                                                                                                                                                                                                                                      | 804 |
| 749 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 805 |
| 750 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 806 |
| 751 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 752 | Zachary Nicholas Houghton, Kenji Sagae, and Emily Morgan. 2025. The role of abstract representations and observed preferences in the ordering of binomials in large language models. In <i>Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)</i> , pages 695–702, Vienna, Austria. Association for Computational Linguistics.                                                                                                                                                                                                                       | 807 |
| 753 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 808 |
| 754 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 809 |
| 755 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 810 |
| 756 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 811 |
| 757 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 812 |
| 758 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 813 |
| 759 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 814 |
| 760 | Zachary Nicholas Houghton, Yu Zhou, Dan Pluth, Jordan Hosier, and Vijay K. Gurbani. 2026. The holistic storage of verb+up phrases in text-based and audio-based language models. <i>Preprint</i> , arXiv:2606.13993.                                                                                                                                                                                                                                                                                                                                                                                               |     |
| 761 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 762 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 763 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 764 | Vsevolod Kapatsinski. 2010. Frequency of use leads to automaticity of production: Evidence from repair in conversation. <i>Language and Speech</i> , 53(1):71–105.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |
| 765 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 766 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 767 | Robert Kubinec. 2023. Ordered beta regression: A parsimonious, well-fitting model for continuous data with lower and upper bounds. <i>Political Analysis</i> , 31(4):519–536.                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |
| 768 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 769 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 770 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 771 | Bob McMurray, Richard N. Aslin, Michael K. Tanenhaus, Michael J. Spivey, and Dana Subik. 2008. Gradient sensitivity to within-category variation in words and syllables. <i>Journal of Experimental Psychology: Human Perception and Performance</i> , 34(6):1609–1631.                                                                                                                                                                                                                                                                                                                                            |     |
| 772 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 773 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 774 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 775 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 776 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 777 | Emily Morgan and Roger Levy. 2015. Modeling idiosyncratic preferences : How generative knowledge and expression frequency jointly determine language structure. pages 1649–1654.                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |
| 778 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 779 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 780 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 781 | Emily Morgan and Roger Levy. 2016. Abstract knowledge versus direct experience in processing of binomial expressions. <i>Cognition</i> , 157:384–402.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
| 782 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 783 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 784 | Emily Morgan and Roger Levy. 2024. Productive knowledge and item-specific knowledge trade off as a function of frequency in multiword expression processing. <i>Language</i> , 100(4):e195–e224.                                                                                                                                                                                                                                                                                                                                                                                                                   |     |
| 785 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 786 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 787 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 788 | Joakim Nivre, Marie-Catherine de Marneffe, Filip Ginter, Jan Hajič, Christopher D. Manning, Sampo Pyysalo, Sebastian Schuster, Francis Tyers, and Daniel Zeman. 2020. Universal Dependencies v2: An evergrowing multilingual treebank collection. In <i>Proceedings of the Twelfth Language Resources and Evaluation Conference</i> , pages 4034–4043, Marseille, France. European Language Resources Association.                                                                                                                                                                                                 |     |
| 789 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 790 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 791 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 792 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 793 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 794 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| 795 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
|     | Dan Pluth, Zachary Nicholas Houghton, Yu Zhou, and Vijay K. Gurbani. 2026. On the interpretability of Whisper encodings using sparse autoencoders. <i>Preprint</i> , arXiv:2605.12225.                                                                                                                                                                                                                                                                                                                                                                                                                             | 796 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 797 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 798 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 799 |
|     | Manuela Sanguinetti and Cristina Bosco. 2015. Part-TUT: The turin university parallel treebank. In <i>Harmonization and Development of Resources and Tools for Italian Natural Language Processing within the PARLI Project</i> , Studies in Computational Intelligence, pages 51–69. Springer International Publishing.                                                                                                                                                                                                                                                                                           | 800 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 801 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 802 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 803 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 804 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 805 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 806 |
|     | Natalia Silveira, Timothy Dozat, Marie-Catherine de Marneffe, Samuel R. Bowman, Miriam Connor, John Bauer, and Chris Manning. 2014. A gold standard dependency corpus for English. In <i>Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC’14)</i> , pages 2897–2904, Reykjavik, Iceland. European Language Resources Association (ELRA).                                                                                                                                                                                                                               | 807 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 808 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 809 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 810 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 811 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 812 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 813 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 814 |
|     | Joseph P. Stemberger and Brian MacWhinney. 2004. Are inflected forms stored in the lexicon. <i>Morphology: Critical concepts in linguistics</i> , 6:107–122.                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 815 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 816 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 817 |
|     | Joseph Paul Stemberger and Brian MacWhinney. 1986. Frequency and the lexical storage of regularly inflected forms. <i>Memory &amp; Cognition</i> , 14(1):17–26.                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 818 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 819 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 820 |
|     | William Timkey and Marten van Schijndel. 2021. All bark and no bite: Rogue dimensions in transformer language models obscure representational quality. In <i>Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)</i> .                                                                                                                                                                                                                                                                                                                                                  | 821 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 822 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 823 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 824 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 825 |
|     | Alex Warstadt, Aaron Mueller, Leshem Choshen, Ethan Wilcox, Chengxu Zhuang, Juan Ciro, Rafael Mosquera, Bhargavi Paranjabe, Adina Williams, Tal Linzen, and Ryan Cotterell. 2023. Findings of the BabyLM challenge: Sample-efficient pretraining on developmentally plausible corpora. In <i>Proceedings of the BabyLM Challenge at CoNLL</i> .                                                                                                                                                                                                                                                                    | 826 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 827 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 828 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 829 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 830 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 831 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 832 |
|     | Amir Zeldes. 2017. The GUM corpus: Creating multilayer resources in the classroom. <i>Language Resources and Evaluation</i> , 51(3):581–612.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 833 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 834 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 835 |
|     | Daniel Zeman, Martin Popel, Milan Straka, Jan Hajič, Joakim Nivre, Filip Ginter, Juhani Luotolahti, Sampo Pyysalo, Slav Petrov, Martin Potthast, Francis Tyers, Elena Badmaeva, Memduh Gokirmak, Anna Nedoluzhko, Silvie Cinková, Jan Hajič jr., Jaroslava Hlaváčová, Václava Kettnerová, Zdeňka Urešová, and 43 others. 2017. CoNLL 2017 shared task: Multilingual parsing from raw text to Universal Dependencies. In <i>Proceedings of the CoNLL 2017 Shared Task: Multilingual Parsing from Raw Text to Universal Dependencies</i> , pages 1–19, Vancouver, Canada. Association for Computational Linguistics. | 836 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 837 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 838 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 839 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 840 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 841 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 842 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 843 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 844 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 845 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 846 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 847 |

A    The Artificial-Language Grid

Table 2 lists every factor crossed in Experiment 1 and its levels. The three effect weights and the context weight define the language; the interaction’s rank is a property of the language too, since it sets how many directions the interaction spans. Hidden width and activation belong to the learner, and the seed redraws both the language’s realization and the model’s initialization.

| factor             | levels              |
|--------------------|---------------------|
| <b>language</b>    |                     |
| item weight        | 0, 0.5, 1, 2        |
| class weight       | 0, 0.5, 1, 2        |
| interaction weight | 0, 0.5, 1, 2        |
| context weight     | 0.5, 2, 6           |
| interaction rank   | 4, 16, 64, 177      |
| <b>learner</b>     |                     |
| hidden width       | 8, 16, 32, 128, 256 |
| activation         | linear, ReLU        |
| <b>replication</b> |                     |
| seed               | 0, 1, 2, 3, 4       |

Table 2: The Experiment 1 grid. The first five rows define an artificial language, the next two the learner, and the last is replication. Item and class ranks are held fixed so that the interaction’s rank is the only one that varies. Crossing every level gives 38,400 cells; the three languages in which all three effect weights are zero contain no item-by-class structure at all and are excluded, leaving the reported total.

Item rank is fixed at 8 and class rank at 4, so the structure a language requires is 12 directions plus the interaction’s rank, running from 16 to 189. Against the widths above, a language needs between 0.062 and 23.6 times as many dimensions as the layer has. Of the 37,800 runs, 99.6% trained to convergence.

## B Validating the Measure

Experiment 1 builds structure into a language and measures a trained model’s representation. However, that approach cannot distinguish a failure of the measure from the model representing the language differently than the structure it was given. Here we remove that ambiguity by constructing representations in which  $\mu$ ,  $\alpha$ ,  $\beta$ , and  $\gamma$  are set directly, with no model or training, and passing the result through the same analysis used for the language models. Ground truth therefore exists at the level where the measure operates.

Specifically, we generated representations with known values of  $\mu$ ,  $\alpha$ ,  $\beta$ , and  $\gamma$ , varying the number of items, classes, representation dimensions, observations per pairing, and noise level across the range used in the experiments. Components were centered to satisfy their defining constraints and rescaled to predetermined shares of the total squared length. The grid included the same factorial Experiment 1 runs, mapped into share space, the three compositions measured in Experiment 2, item counts from 27 to 180, two and four classes, widths from 8 to 2048, 10, 24, and 100 observations per item and class, three noise levels, and five seeds. We also generated orthogonal and known non-orthogonal item and class effects: the orthogonal condition draws  $\alpha$  and  $\beta$  independently, while the non-orthogonal condition places a known fraction of  $\beta$  along  $\alpha$ ’s leading singular direction. In total, this gave 252,000 runs.

At 24 observations per item and class, the measured interaction matches the one it was given across the whole range (Table 3), and is flagged in essentially every run where anything was present and about one in twenty where nothing was.

| specified | runs | measured                | flagged |
|-----------|------|-------------------------|---------|
| 0.000     | 6300 | 0.0000 [0.0000, 0.0094] | 5%      |
| 0.030     | 900  | 0.0323 [0.0199, 0.0418] | 100%    |
| 0.048     | 1800 | 0.0495 [0.0385, 0.0688] | 100%    |
| 0.056     | 1800 | 0.0573 [0.0458, 0.0918] | 100%    |
| 0.059     | 1800 | 0.0607 [0.0495, 0.1041] | 100%    |
| 0.100     | 1800 | 0.1007 [0.0911, 0.1089] | 100%    |
| 0.111     | 900  | 0.1154 [0.1031, 0.1279] | 100%    |
| 0.167     | 1800 | 0.1708 [0.1594, 0.2204] | 100%    |
| 0.190     | 1800 | 0.1937 [0.1825, 0.2736] | 100%    |
| 0.200     | 1800 | 0.2030 [0.1926, 0.2981] | 100%    |
| 0.240     | 900  | 0.2416 [0.2327, 0.2488] | 100%    |
| 0.333     | 900  | 0.3396 [0.3263, 0.3610] | 100%    |
| 0.444     | 1800 | 0.4488 [0.4381, 0.5059] | 100%    |
| 0.485     | 1800 | 0.4877 [0.4787, 0.5645] | 100%    |
| 0.500     | 1800 | 0.5025 [0.4941, 0.5873] | 100%    |
| 0.667     | 900  | 0.6702 [0.6606, 0.6814] | 100%    |
| 0.762     | 1800 | 0.7634 [0.7551, 0.7828] | 100%    |
| 0.800     | 1800 | 0.8009 [0.7931, 0.8242] | 100%    |
| 0.889     | 900  | 0.8895 [0.8809, 0.8972] | 100%    |
| 0.941     | 1800 | 0.9412 [0.9339, 0.9471] | 100%    |
| 1.000     | 900  | 0.9999 [0.9936, 1.0000] | 100%    |

Table 3: Recovery at 24 observations per item and class in 768 or more dimensions, pooled over noise levels, item counts, and widths. Brackets are the 2.5th and 97.5th percentiles across runs.

All three effects read essentially zero when a component is set to zero, at every sample size, and each is flagged at approximately the nominal 5% rate (Table 4).

| effect       | observations | median  | 95th pct | flagged |
|--------------|--------------|---------|----------|---------|
| class effect | 10           | 0.00000 | 0.0081   | 6.1%    |
| class effect | 24           | 0.00000 | 0.0034   | 6.5%    |
| class effect | 100          | 0.00000 | 0.0009   | 6.3%    |
| interaction  | 10           | 0.00009 | 0.0644   | 6.2%    |
| interaction  | 24           | 0.00004 | 0.0309   | 5.9%    |
| interaction  | 100          | 0.00002 | 0.0083   | 4.9%    |
| item effect  | 10           | 0.00007 | 0.0568   | 4.1%    |
| item effect  | 24           | 0.00003 | 0.0255   | 3.6%    |
| item effect  | 100          | 0.00000 | 0.0064   | 5.0%    |

Table 4: What the measure reports when an effect is set to exactly zero, by observations per item and class, pooled over every other axis of the validation grid.

The false-positive rate varies somewhat with representation width and component, particularly at small widths, but remains below 10% throughout the validation grid except for the class effect at the narrowest width, where it reaches 14% (Table 5). The language-model analyses use  $d \geq 768$ , where the rates are closer to the nominal level.

| width d | item | class | interaction |
|---------|------|-------|-------------|
| 8       | 3%   | 14%   | 7%          |
| 16      | 7%   | 3%    | 5%          |
| 32      | 4%   | 4%    | 10%         |
| 128     | 6%   | 4%    | 5%          |
| 768     | 2%   | 8%    | 3%          |
| 1024    | 3%   | 5%    | 3%          |
| 2048    | 0%   | 8%    | 8%          |

Table 5: How often a component set to exactly zero is flagged, by width and by effect.

| width d | measured                   | analytic chance |
|---------|----------------------------|-----------------|
| 8       | 0.17501 [0.06900, 0.40018] | 0.25000         |
| 16      | 0.10678 [0.03744, 0.20884] | 0.12500         |
| 32      | 0.05609 [0.01864, 0.10454] | 0.06250         |
| 128     | 0.01714 [0.00450, 0.02611] | 0.02344         |
| 768     | 0.00261 [0.00079, 0.00472] | 0.00391         |
| 1024    | 0.00217 [0.00065, 0.00330] | 0.00293         |
| 2048    | 0.00096 [0.00031, 0.00165] | 0.00146         |

Table 6: Measured item-class overlap when the two effects are constructed orthogonal, at 24 observations per item and class. *Measured* is the median over the runs at that width with the 2.5th and 97.5th percentiles in brackets. The *chance* column is the analytic level of Appendix C.

With orthogonal item and class effects, the measured overlap tracks the chance level across representation widths (Table 6). When a known fraction of the class direction is placed inside the item subspace, the measured overlap rises above chance and is detected in 89% of runs at  $d = 2048$  (Table 7). Thus, the measure recovers both the expected chance overlap under orthogonality and genuine non-orthogonality.

| width d | measured                | chance  | detected |
|---------|-------------------------|---------|----------|
| 8       | 0.1688 [0.0824, 0.3998] | 0.12500 | 9%       |
| 16      | 0.0951 [0.0414, 0.2026] | 0.06250 | 22%      |
| 32      | 0.0593 [0.0225, 0.1119] | 0.03125 | 36%      |
| 128     | 0.0213 [0.0062, 0.0517] | 0.00781 | 63%      |
| 768     | 0.0065 [0.0011, 0.0298] | 0.00260 | 83%      |
| 1024    | 0.0056 [0.0008, 0.0331] | 0.00098 | 82%      |
| 2048    | 0.0036 [0.0004, 0.0290] | 0.00049 | 89%      |

Table 7: Measured item-class overlap when a fraction of the class direction is placed **INSIDE** the item subspace (non-orthogonal), at 24 observations per item and class. *Measured* is the median over the runs at that width with the 2.5th and 97.5th percentiles in brackets; *chance* is the analytic random-orientation level; *detected* is the fraction of runs whose overlap falls outside that null’s 95% interval.

## B.1 Within-cell spread

The interaction is a residual, and a residual absorbs whatever noise the terms before it do not. Each cell mean is an average over the contexts a word appears in, so it carries sampling error of order the within-cell spread over the number of tokens; estimated within a single split, the interaction would inherit that error and grow with the spread whether or not any interaction was present.

The cross-split product forecloses this rather than reducing it. The two estimates come from disjoint halves of a cell’s tokens, so their errors are independent and mean-zero, and the expectation of their inner product is the true squared length with no variance term: the spread enters the variance of the estimate and not its expectation. This holds for any spread, and for a spread that differs from cell to cell.

Table ef{tbl-val-noise} shows it. Holding the planted interaction fixed and scaling the within-cell spread by fifty leaves the measured value where it is, and where nothing was planted the false-positive rate stays at the nominal level rather than rising with the spread. If anything the measure drifts slightly low at the widest spread.

| specified | noise | runs | measured | flagged |
|-----------|-------|------|----------|---------|
| 0.00      | 1x    | 6300 | 0.0000   | 6%      |
|           | 10x   | 6300 | 0.0000   | 6%      |
|           | 50x   | 6300 | 0.0001   | 6%      |
| 0.10      | 1x    | 1800 | 0.1012   | 100%    |
|           | 10x   | 1800 | 0.1002   | 100%    |
|           | 50x   | 1800 | 0.1001   | 100%    |
| 0.20      | 1x    | 1800 | 0.2199   | 100%    |
|           | 10x   | 1800 | 0.2049   | 100%    |
|           | 50x   | 1800 | 0.2016   | 100%    |

Table 8: Recovery against within-cell spread, in 768 or more dimensions, pooled over item counts, widths and sample sizes. *Noise* is the multiple of the baseline within-cell spread the representations were generated with. *Measured* is the median across runs and *flagged* the share whose re-split interval excludes zero, which should sit at 5% in the rows where nothing was planted.

Two things outside the validation grid point the same way. The independence the argument rests on would fail if a cell’s tokens were not independent draws, the same word in the same class pulled repeatedly from one passage; across the items measured here, 2.6% of occurrences share a sentence with another occurrence of the same word in the same class, and the most affected item reaches 15%. And in Experiment 1, where the spread is manipulated directly rather than observed, raising the context weight *lowers* the measured interaction (Table ef{tbl-lang}), which is the opposite of what an inflation account predicts.

## C The Decomposition

### C.1 The item-by-class means

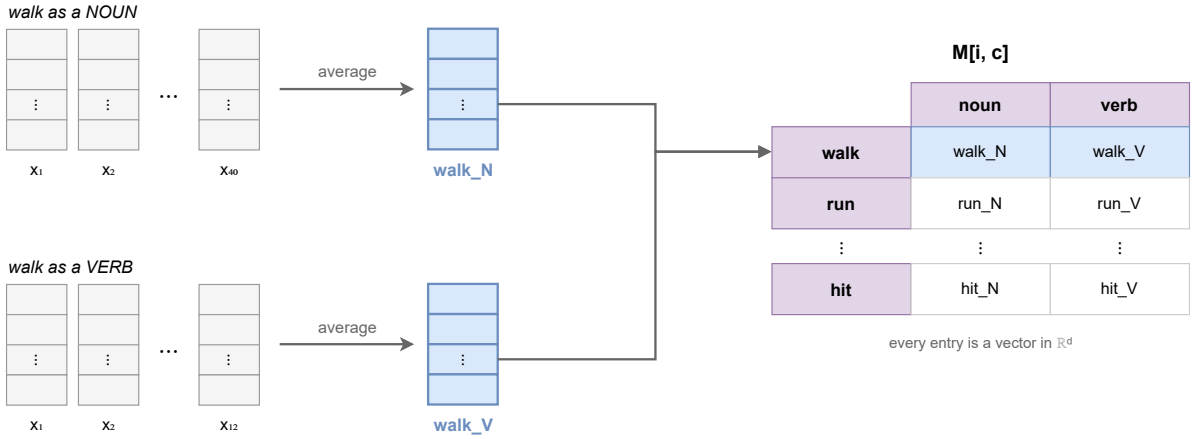


Figure 5: Each word in each class is observed in many contexts, and those observations are averaged within the pairing to give one of the item-by-class means. Every one of them is a vector in  $\mathbb{R}^d$ , and every item appears in every class.

A word used in a given class is observed in many contexts, and the representation differs a little in each. Averaging those observations within the pairing gives one of the item-by-class means: for items  $i \in \{1, \dots, I\}$  crossed with classes  $c \in \{1, \dots, C\}$ ,  $M[i, c] \in \mathbb{R}^d$  is the mean representation of item  $i$  in class  $c$  (Figure 5). An item observed in only one class is dropped, as is an item with fewer than 10 tokens in either class, so the set is complete and every mean rests on a comparable number of observations. Because every item appears at every class, the means arrange into  $I$  by  $C$  positions, and the rest of this appendix calls that arrangement the grid when what matters is the positions rather than the vectors sitting at them. The variation across contexts that the averaging sets aside is not discarded; it is what the two splits of Section 2.2.2 are drawn from.

### C.2 Sizes

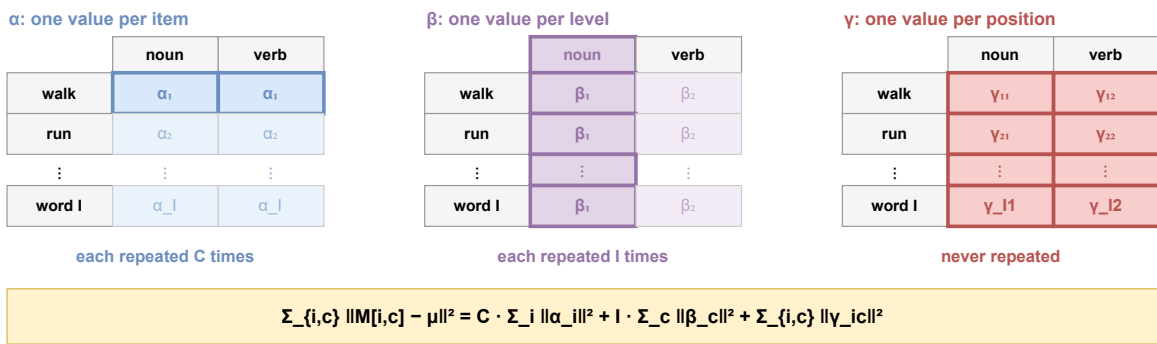


Figure 6: How many positions of the grid each component occupies.  $\alpha_i$  does not depend on the class, so it occupies all  $C$  positions of its row;  $\beta_c$  does not depend on the item, so it occupies all  $I$  positions of its column;  $\gamma_{ic}$  takes its own value at every position and is never repeated.

Since the effects are centered along their respective axes, their cross-terms cancel when summed over the full grid. Thus the total squared length partitions exactly:

$$\sum_{i,c} \|M[i, c] - \mu\|^2 = C \sum_i \|\alpha_i\|^2 + I \sum_c \|\beta_c\|^2 + \sum_{i,c} \|\gamma_{ic}\|^2,$$

Sizes are reported as shares of the total. All three terms scale linearly in  $I$ , so the shares do not depend on how many items there are and are comparable across categories of different size.

The above decomposition, however, ignores the role of context. In order to estimate how the three components and context vary in size, we can instead decompose each representation into its pairing mean and its deviation from that mean, giving the standard within–between partition:

$$\sum_x \|x - \bar{x}\|^2 = \underbrace{\sum_{i,c} \sum_{x \in (i,c)} \|x - \hat{M}[i, c]\|^2}_{\text{within a pairing}} + \underbrace{\sum_{i,c} n_{ic} \|\hat{M}[i, c] - \bar{x}\|^2}_{\text{between pairings}}.$$

The between-pairing term is the variation partitioned into item, class, and item-by-class components above, while the within-pairing term captures variation across contexts. The resulting context variance and the  $\alpha$ ,  $\beta$ , and  $\gamma$  variance components together partition the variance of the original representations.

### C.3 The overlaps and their chance level

The component sizes describe how much variation each component accounts for, but not whether the components occupy the same directions in  $\mathbb{R}^d$ . To measure their overlap, we project each component into a target subspace and ask how much of its squared length is retained. Specifically, we project  $\alpha$  into the subspace spanned by  $\beta$ ,  $\beta$  into the subspace spanned by  $\alpha$ , and  $\gamma$  into the subspace jointly spanned by  $\alpha$  and  $\beta$ . If  $S$  is an orthonormal basis for the target subspace and  $V$  is the component being projected, the overlap is

$$\frac{\|VS\|_F^2}{\|V\|_F^2}$$

Thus, an overlap of zero means that the component is orthogonal to the target subspace, whereas an overlap of one means that it lies entirely within it. Because the projection uses squared lengths, opposite directions contribute equally rather than canceling. We use the leading singular vectors of each target component to define its subspace, with the number of directions determined by its participation ratio. The number of directions in the target subspace therefore determines its chance overlap under random orientation.

The chance level for overlap is  $r/d$ , where  $r$  is the dimensionality of the target subspace and  $d$  is the full representation dimension. A randomly oriented component therefore has an expected fraction  $r/d$  of its squared length in the target subspace. At  $d = 2048$ , for example, the chance level is 0.0005 for a one-dimensional class subspace. We therefore interpret observed overlaps relative to this chance level.

## D Width and Orthogonality

Whether the item and class effects are orthogonal depends on the width of the representation. Figure 9 reports that relationship from the fitted models; this appendix gives the remaining properties of the artificial-language runs.

Under ReLU, overlap stays below chance at wide layers and rises above it as the layer narrows, while the linear condition stays well below chance at every width. Varying the rank of the interaction moves that crossing only slightly, and far less than it moves the interaction’s own overlap with the margins.

At low widths, the representations also vary substantially across runs, with this variability decreasing as width increases. For example, two runs of the same language differ by 0.70 in the item effect and 0.73 in the class effect at width 8, compared with 0.16 and 0.15 at width 32. This variability reflects the different ways models can distribute information when representational capacity is limited.

Two properties of the language also affect the composition of the item-by-class structure (Table 9). Increasing the contribution of context reduces the interaction share, while increasing the item’s share in the language produces a corresponding increase in its share of the representation.

| property                | level        | runs | item  | class | interaction |
|-------------------------|--------------|------|-------|-------|-------------|
| context in the language | under 30%    | 3600 | 0.490 | 0.399 | 0.028       |
|                         | 30 to 70%    | 2871 | 0.538 | 0.371 | 0.008       |
|                         | over 70%     | 3886 | 0.578 | 0.350 | 0.004       |
| item against class      | item < class | 3525 | 0.428 | 0.480 | 0.017       |
|                         | item = class | 3427 | 0.569 | 0.359 | 0.015       |
|                         | item > class | 3405 | 0.644 | 0.294 | 0.011       |

Table 9: How two other properties of the artificial language move the composition of the item-by-class structure, under ReLU, pooled over widths. Rows within a property partition the same runs. Values are medians of the three shares over the runs in the row.

Whether an interaction is detected at all, rather than how large it is, separates the two activations further. Every language counted here holds both an item and a class effect, so an additive one is genuinely an item part plus a class part. Across the 8,100 of them that also contained an interaction, the linear learner’s interaction interval excludes zero in 50% of them against 70% for ReLU; where the language contained no interaction the rates were 5% and 30% respectively, against a test that flags a truly-zero component about 5% of the time (Appendix B). A linear learner therefore builds interaction structure it has no parameter for, and a ReLU learner registers one even where the language holds none.

## D.1 Component models

Table 10 reports every coefficient of Equation 3 and Figure 7 shows them.

We fit each activation separately, giving six models in all, so that the linear arm is described by the same model as the ReLU arm rather than by medians alone. We keep only runs that converged and in which at least one of the three components has a re-split interval excluding zero. That second criterion matters more here than it does for the descriptive figure: where nothing resolves, each of the three shares is a noise-sized quantity divided by a noise-sized total, so whichever component happens to land slightly positive takes nearly all of it, and the resulting value is arbitrary rather than small. Unlike the descriptive figure, we do keep languages in which a component was given no weight at all, since those runs anchor the low end of the effect-size predictors and are what identify the off-diagonal terms.

Every component and overlap model used at least 36,000 post-warmup draws, produced no divergent transitions, and reached a bulk effective sample size of at least 1,556. The largest  $\hat{R}$  across them is 1.01.

Because Equation 3 carries a three-way interaction, the width and rank coefficients are conditional main effects: each is the effect at the mean of the other predictors rather than an effect averaged over them. The contrast in Table 11 is evaluated at the same point.

Figure 10 holds rank at its mean, which is the right simplification for the main text but leaves the three-way term invisible. Figure 8 shows the same fitted surface with rank broken out as a row of facets, so the question of whether the rank an interaction requires constrains what the model represents can be read directly: were it to constrain it, the rows would differ, most visibly in the interaction column.

| arm    | component   | term                       | estimate | error | 95% CI         | P(b > 0) |
|--------|-------------|----------------------------|----------|-------|----------------|----------|
| relu   | class       | intercept                  | -0.45    | 0.04  | [-0.53, -0.38] | 0.00     |
|        | class       | item                       | -1.47    | 0.16  | [-1.79, -1.15] | 0.00     |
|        | class       | class                      | 2.01     | 0.17  | [1.67, 2.35]   | 1.00     |
|        | class       | interaction                | 0.29     | 0.17  | [-0.03, 0.62]  | 0.96     |
|        | class       | width                      | 0.03     | 0.00  | [0.02, 0.04]   | 1.00     |
|        | class       | rank                       | -0.05    | 0.03  | [-0.10, 0.01]  | 0.05     |
|        | class       | width x rank               | -0.01    | 0.00  | [-0.02, -0.01] | 0.00     |
|        | class       | effect size x width        | -0.68    | 0.02  | [-0.71, -0.64] | 0.00     |
|        | class       | effect size x rank         | 0.03     | 0.12  | [-0.22, 0.27]  | 0.58     |
|        | class       | effect size x width x rank | -0.03    | 0.01  | [-0.06, -0.00] | 0.02     |
|        | class       | sd(language)               | 0.98     | 0.03  | [0.93, 1.04]   | 1.00     |
|        | class       | phi                        | 3.77     | 0.04  | [3.70, 3.85]   | 1.00     |
|        | class       | cutzero                    | -4.76    | 0.05  | [-4.86, -4.65] | 0.00     |
|        | class       | cutone                     | 2.22     | 0.01  | [2.20, 2.23]   | 1.00     |
| linear | class       | intercept                  | -0.99    | 0.05  | [-1.09, -0.89] | 0.00     |
|        | class       | item                       | -3.04    | 0.22  | [-3.48, -2.60] | 0.00     |
|        | class       | class                      | 4.10     | 0.20  | [3.71, 4.50]   | 1.00     |
|        | class       | interaction                | -1.13    | 0.22  | [-1.56, -0.69] | 0.00     |
|        | class       | width                      | -0.23    | 0.00  | [-0.24, -0.23] | 0.00     |
|        | class       | rank                       | -0.01    | 0.04  | [-0.08, 0.07]  | 0.44     |
|        | class       | width x rank               | -0.01    | 0.00  | [-0.01, -0.00] | 0.00     |
|        | class       | effect size x width        | -0.47    | 0.01  | [-0.50, -0.45] | 0.00     |
|        | class       | effect size x rank         | 0.03     | 0.14  | [-0.26, 0.30]  | 0.58     |
|        | class       | effect size x width x rank | -0.03    | 0.01  | [-0.05, -0.01] | 0.00     |
|        | class       | sd(language)               | 1.41     | 0.04  | [1.34, 1.49]   | 1.00     |
|        | class       | phi                        | 15.06    | 0.17  | [14.72, 15.40] | 1.00     |
|        | class       | cutzero                    | -5.66    | 0.05  | [-5.75, -5.57] | 0.00     |
|        | class       | cutone                     | 2.37     | 0.01  | [2.36, 2.39]   | 1.00     |
| relu   | interaction | intercept                  | -3.33    | 0.03  | [-3.39, -3.28] | 0.00     |
|        | interaction | item                       | -0.17    | 0.12  | [-0.40, 0.05]  | 0.07     |
|        | interaction | class                      | 0.12     | 0.12  | [-0.12, 0.35]  | 0.84     |
|        | interaction | interaction                | 2.60     | 0.12  | [2.37, 2.83]   | 1.00     |
|        | interaction | width                      | 0.57     | 0.00  | [0.56, 0.58]   | 1.00     |
|        | interaction | rank                       | -0.02    | 0.02  | [-0.05, 0.02]  | 0.20     |
|        | interaction | width x rank               | 0.04     | 0.00  | [0.03, 0.04]   | 1.00     |
|        | interaction | effect size x width        | -0.19    | 0.01  | [-0.22, -0.17] | 0.00     |
|        | interaction | effect size x rank         | 0.10     | 0.08  | [-0.06, 0.27]  | 0.89     |
|        | interaction | effect size x width x rank | -0.06    | 0.01  | [-0.07, -0.04] | 0.00     |
|        | interaction | sd(language)               | 0.69     | 0.02  | [0.66, 0.74]   | 1.00     |
|        | interaction | phi                        | 26.14    | 0.36  | [25.43, 26.86] | 1.00     |
|        | interaction | cutzero                    | -5.18    | 0.03  | [-5.23, -5.13] | 0.00     |
|        | interaction | cutone                     | 2.29     | 0.03  | [2.24, 2.35]   | 1.00     |
| linear | interaction | intercept                  | -4.60    | 0.03  | [-4.65, -4.55] | 0.00     |
|        | interaction | item                       | 0.21     | 0.10  | [0.03, 0.41]   | 0.99     |
|        | interaction | class                      | -0.44    | 0.11  | [-0.65, -0.24] | 0.00     |
|        | interaction | interaction                | 2.07     | 0.10  | [1.88, 2.26]   | 1.00     |
|        | interaction | width                      | 0.19     | 0.00  | [0.18, 0.20]   | 1.00     |
|        | interaction | rank                       | -0.08    | 0.02  | [-0.11, -0.05] | 0.00     |
|        | interaction | width x rank               | 0.03     | 0.00  | [0.02, 0.04]   | 1.00     |
|        | interaction | effect size x width        | 0.36     | 0.02  | [0.33, 0.39]   | 1.00     |
|        | interaction | effect size x rank         | 0.21     | 0.07  | [0.07, 0.34]   | 1.00     |
|        | interaction | effect size x width x rank | 0.03     | 0.01  | [0.01, 0.05]   | 1.00     |
|        | interaction | sd(language)               | 0.55     | 0.02  | [0.50, 0.59]   | 1.00     |
|        | interaction | phi                        | 37.86    | 0.62  | [36.65, 39.08] | 1.00     |
|        | interaction | cutzero                    | -5.22    | 0.02  | [-5.26, -5.17] | 0.00     |
|        | interaction | cutone                     | 2.26     | 0.05  | [2.16, 2.37]   | 1.00     |
| relu   | item        | intercept                  | 0.22     | 0.04  | [0.15, 0.29]   | 1.00     |
|        | item        | item                       | 1.59     | 0.16  | [1.28, 1.89]   | 1.00     |
|        | item        | class                      | -1.96    | 0.17  | [-2.29, -1.62] | 0.00     |
|        | item        | interaction                | -0.45    | 0.16  | [-0.76, -0.14] | 0.00     |
|        | item        | width                      | -0.09    | 0.00  | [-0.10, -0.09] | 0.00     |
|        | item        | rank                       | 0.02     | 0.03  | [-0.03, 0.07]  | 0.82     |
|        | item        | width x rank               | 0.01     | 0.00  | [-0.00, 0.01]  | 0.97     |
|        | item        | effect size x width        | -0.12    | 0.02  | [-0.15, -0.09] | 0.00     |
|        | item        | effect size x rank         | -0.14    | 0.11  | [-0.36, 0.08]  | 0.11     |
|        | item        | effect size x width x rank | -0.04    | 0.01  | [-0.07, -0.02] | 0.00     |
|        | item        | sd(language)               | 0.95     | 0.03  | [0.90, 1.01]   | 1.00     |
|        | item        | phi                        | 3.38     | 0.03  | [3.31, 3.45]   | 1.00     |
|        | item        | cutzero                    | -3.84    | 0.05  | [-3.93, -3.74] | 0.00     |
|        | item        | cutone                     | 2.19     | 0.01  | [2.18, 2.21]   | 1.00     |
| linear | item        | intercept                  | 0.92     | 0.05  | [0.82, 1.03]   | 1.00     |
|        | item        | item                       | 2.84     | 0.23  | [2.38, 3.30]   | 1.00     |
|        | item        | class                      | -3.77    | 0.20  | [-4.17, -3.37] | 0.00     |
|        | item        | interaction                | 1.23     | 0.23  | [0.77, 1.68]   | 1.00     |
|        | item        | width                      | 0.25     | 0.00  | [0.24, 0.25]   | 1.00     |
|        | item        | rank                       | -0.00    | 0.04  | [-0.08, 0.07]  | 0.46     |
|        | item        | width x rank               | 0.01     | 0.00  | [0.00, 0.01]   | 1.00     |
|        | item        | effect size x width        | -0.72    | 0.01  | [-0.74, -0.70] | 0.00     |
|        | item        | effect size x rank         | 0.01     | 0.17  | [-0.32, 0.33]  | 0.52     |
|        | item        | effect size x width x rank | -0.02    | 0.01  | [-0.04, -0.00] | 0.01     |
|        | item        | sd(language)               | 1.47     | 0.04  | [1.39, 1.55]   | 1.00     |

| arm    | component   | estimate | error | 95% CI         | excludes zero |
|--------|-------------|----------|-------|----------------|---------------|
| relu   | class       | -0.02    | 0.03  | [-0.07, 0.04]  | no            |
| linear | class       | -0.24    | 0.04  | [-0.32, -0.16] | yes           |
| relu   | interaction | 0.55     | 0.02  | [0.51, 0.59]   | yes           |
| linear | interaction | 0.10     | 0.02  | [0.07, 0.14]   | yes           |
| relu   | item        | -0.07    | 0.03  | [-0.12, -0.02] | yes           |
| linear | item        | 0.24     | 0.04  | [0.16, 0.32]   | yes           |

Table 11: The sum of the width and rank coefficients, for each measured component and activation. A representation limited by how many dimensions its structure needs relative to the layer would give width and rank equal and opposite coefficients, so a sum whose interval excludes zero is evidence against that account.

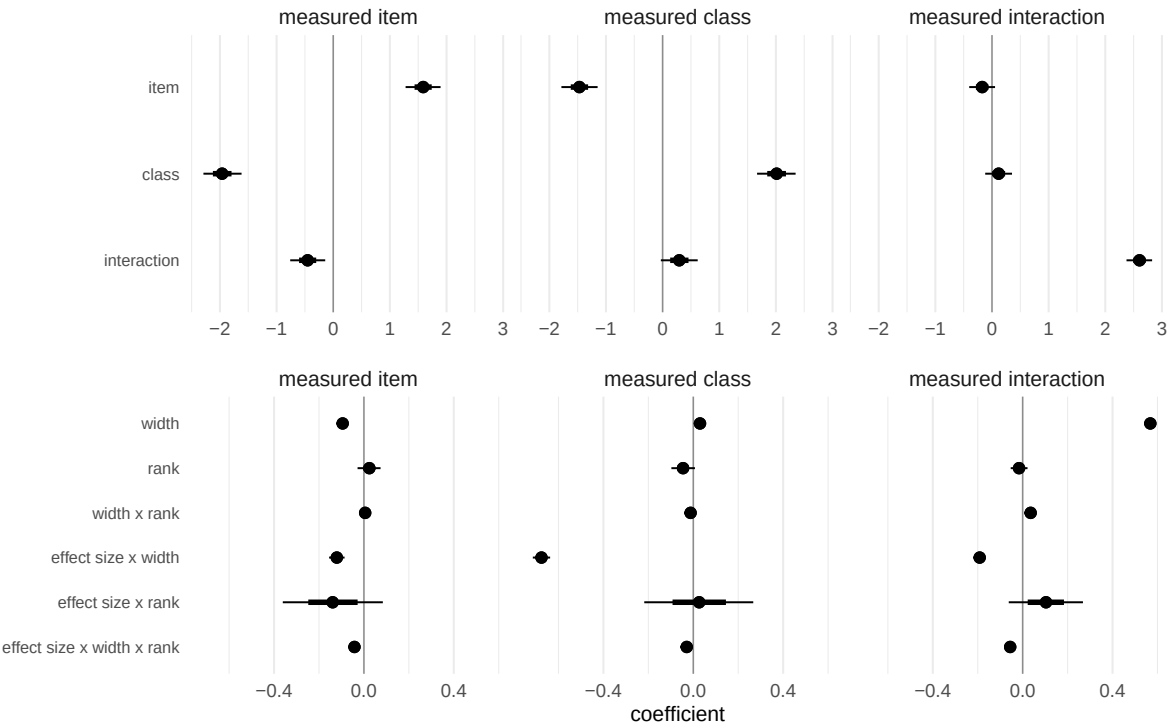


Figure 7: Every coefficient of Equation 3, with 66% and 95% posterior intervals. Columns are the three measured components, one fitted model each. The upper block is the three effect sizes in the language, whose diagonal is the specificity check; the lower block is the architecture terms, where *effect size* refers to the panel’s own component. The two blocks are drawn on separate x scales, since the language terms span a range several times wider than the architecture terms.

D.2 Overlap models

Table 12 reports every coefficient of Equation 4. There are four of these models, two measures crossed with the two activations, and they differ from the component models in what the architecture is crossed with: an overlap is a property of two components together, so there is no single “own” effect size to interact with width and rank, and the three effect sizes enter additively as controls instead.

The inclusion rule is the one the component models use. A run contributes if it converged and if at least one of the three components has a re-split interval excluding zero; where none does, the decomposition is noise throughout, and the directions one would project between are as arbitrary as the shares. Within that set nothing is excluded on the size of the components an overlap relates, so a component too small to orient contributes the overlap it produces, which is the value an arbitrary direction would give and is what the random-subspace baseline is there to reveal.

The outcome is the overlap divided by its own chance level and logged, so a coefficient is a multiplicative effect on that ratio and zero means no change in it. The chance level is that run’s own random-subspace

1000  
1001  
1002

median, which is what makes widths comparable at all: the same raw overlap means different things at 8 dimensions and at 2048, and the median tracks the rank of the target subspace over the width closely enough that the ratio reads as a multiple of what an arbitrary direction would give.

| arm    | measure     | term         | estimate | error | 95% CI         | P(b > 0) |
|--------|-------------|--------------|----------|-------|----------------|----------|
| relu   | int_margins | intercept    | 0.49     | 0.01  | [0.47, 0.51]   | 1.00     |
|        | int_margins | item         | -1.00    | 0.05  | [-1.10, -0.92] | 0.00     |
|        | int_margins | class        | -0.64    | 0.05  | [-0.74, -0.55] | 0.00     |
|        | int_margins | interaction  | -0.65    | 0.05  | [-0.74, -0.56] | 0.00     |
|        | int_margins | width        | -0.32    | 0.00  | [-0.32, -0.31] | 0.00     |
|        | int_margins | rank         | -0.04    | 0.01  | [-0.05, -0.02] | 0.00     |
|        | int_margins | width x rank | -0.02    | 0.00  | [-0.03, -0.02] | 0.00     |
|        | int_margins | sd(language) | 0.23     | 0.01  | [0.21, 0.25]   | 1.00     |
|        | int_margins | sigma        | 0.79     | 0.00  | [0.78, 0.80]   | 1.00     |
| linear | int_margins | intercept    | -0.53    | 0.02  | [-0.56, -0.49] | 0.00     |
|        | int_margins | item         | -1.75    | 0.08  | [-1.92, -1.59] | 0.00     |
|        | int_margins | class        | -0.69    | 0.09  | [-0.86, -0.51] | 0.00     |
|        | int_margins | interaction  | -0.44    | 0.08  | [-0.60, -0.28] | 0.00     |
|        | int_margins | width        | 0.10     | 0.00  | [0.10, 0.11]   | 1.00     |
|        | int_margins | rank         | -0.05    | 0.01  | [-0.07, -0.02] | 0.00     |
|        | int_margins | width x rank | -0.02    | 0.00  | [-0.03, -0.02] | 0.00     |
|        | int_margins | sd(language) | 0.49     | 0.01  | [0.46, 0.52]   | 1.00     |
|        | int_margins | sigma        | 0.73     | 0.00  | [0.72, 0.74]   | 1.00     |
| relu   | item_class  | intercept    | 0.33     | 0.01  | [0.30, 0.36]   | 1.00     |
|        | item_class  | item         | -0.51    | 0.07  | [-0.65, -0.38] | 0.00     |
|        | item_class  | class        | -1.80    | 0.07  | [-1.95, -1.66] | 0.00     |
|        | item_class  | interaction  | 0.23     | 0.07  | [0.10, 0.36]   | 1.00     |
|        | item_class  | width        | -0.40    | 0.00  | [-0.41, -0.39] | 0.00     |
|        | item_class  | rank         | -0.03    | 0.01  | [-0.05, -0.01] | 0.00     |
|        | item_class  | width x rank | -0.01    | 0.00  | [-0.02, -0.01] | 0.00     |
|        | item_class  | sd(language) | 0.37     | 0.01  | [0.34, 0.39]   | 1.00     |
|        | item_class  | sigma        | 0.90     | 0.00  | [0.89, 0.91]   | 1.00     |
| linear | item_class  | intercept    | -1.05    | 0.01  | [-1.08, -1.03] | 0.00     |
|        | item_class  | item         | -0.58    | 0.05  | [-0.69, -0.47] | 0.00     |
|        | item_class  | class        | -0.48    | 0.06  | [-0.59, -0.36] | 0.00     |
|        | item_class  | interaction  | 0.11     | 0.05  | [0.01, 0.22]   | 0.98     |
|        | item_class  | width        | -0.00    | 0.00  | [-0.01, 0.00]  | 0.22     |
|        | item_class  | rank         | -0.05    | 0.01  | [-0.06, -0.03] | 0.00     |
|        | item_class  | width x rank | -0.02    | 0.00  | [-0.02, -0.01] | 0.00     |
|        | item_class  | sd(language) | 0.31     | 0.01  | [0.29, 0.33]   | 1.00     |
|        | item_class  | sigma        | 0.64     | 0.00  | [0.64, 0.65]   | 1.00     |

Table 12: Every coefficient of Equation 4, in one block per measure and activation. *Estimate* is the posterior mean and *error* its standard deviation; the interval runs from the 2.5th to the 97.5th percentile of the posterior. The last column is the share of posterior draws above zero. The three effect sizes enter as additive controls, and the architecture terms are width, rank and their interaction.

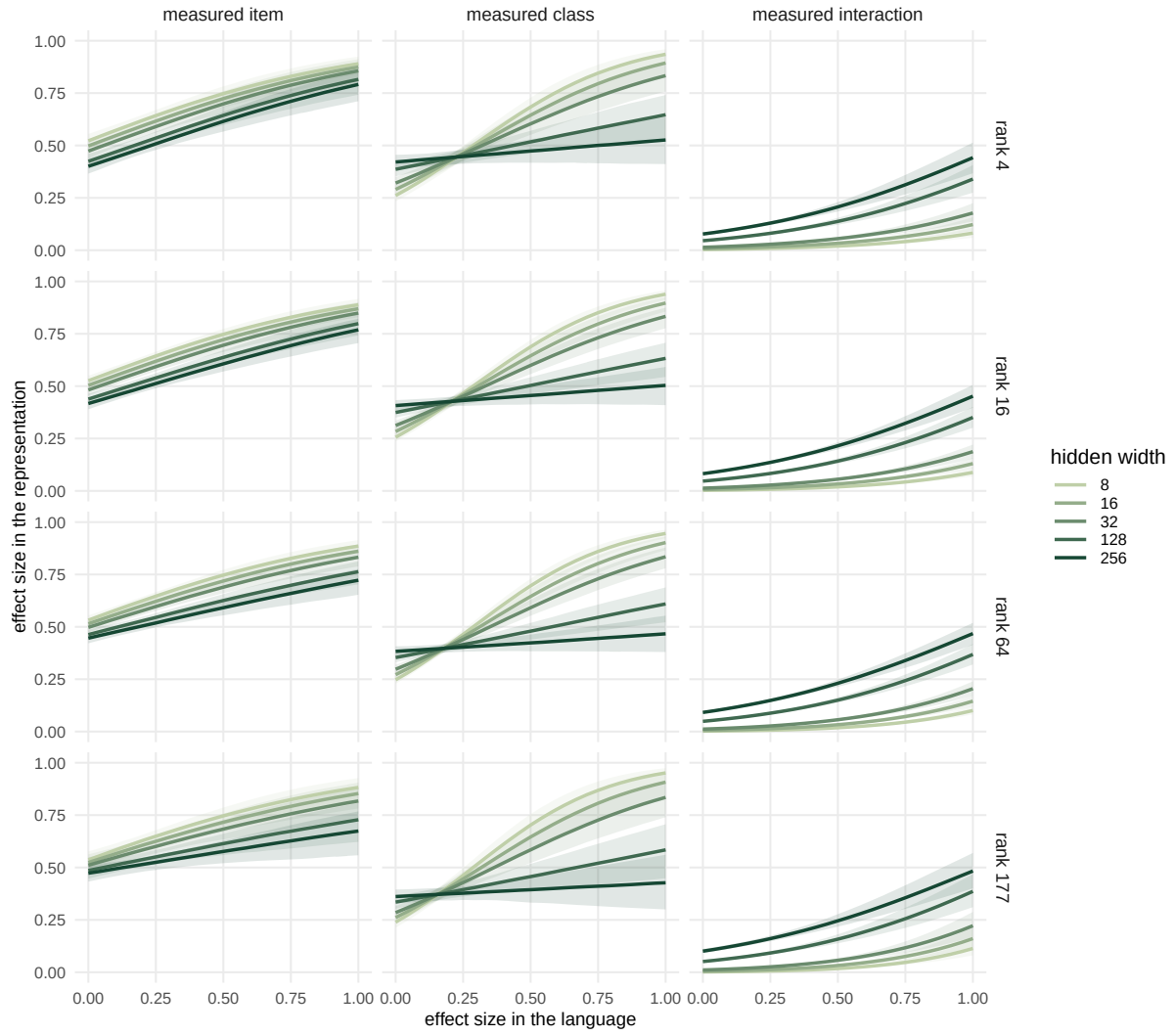


Figure 8: The fitted surface of Equation 3 with the rank of the interaction in the language broken out as rows, rather than held at its mean as in Figure 10. Columns are the three measured components and lines are hidden widths, with 95% intervals. Rows that look alike indicate that the rank an interaction requires does not constrain what the learner represents of it.

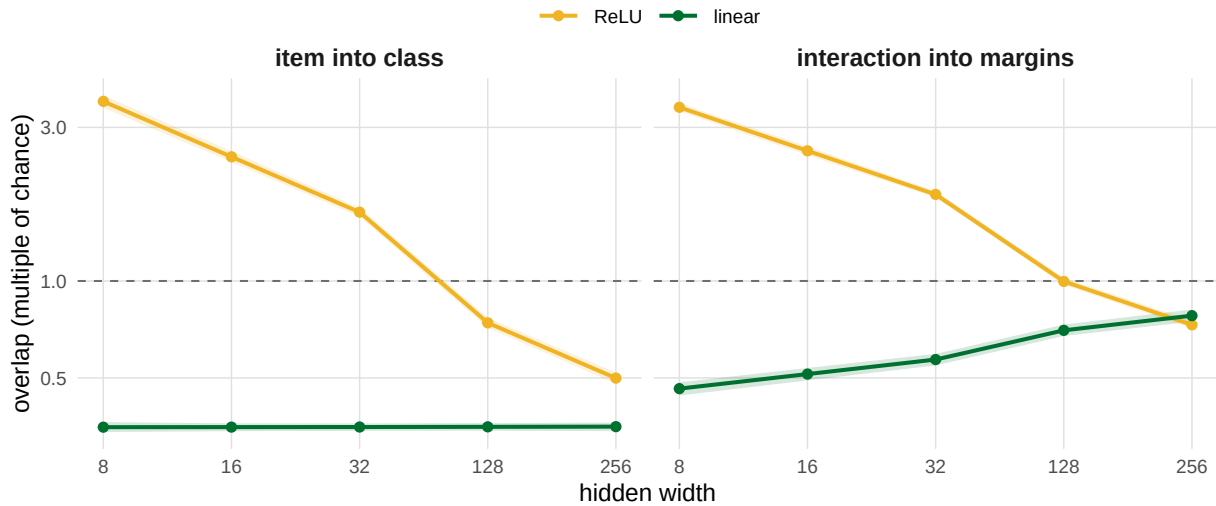


Figure 9: Fitted overlap between components, as a multiple of the level an arbitrary orientation would give, against hidden width. Left: the item effect projected onto the class direction. Right: the interaction projected onto the span of both margins. The dashed line is chance; above it components share directions more than orientation alone explains, below it the learner has separated them by more than chance. Widths are drawn evenly spaced rather than to scale, since width 64 was not trained. Effect sizes are held at their means, and every run contributes: no overlap is excluded on the size of the components it relates.

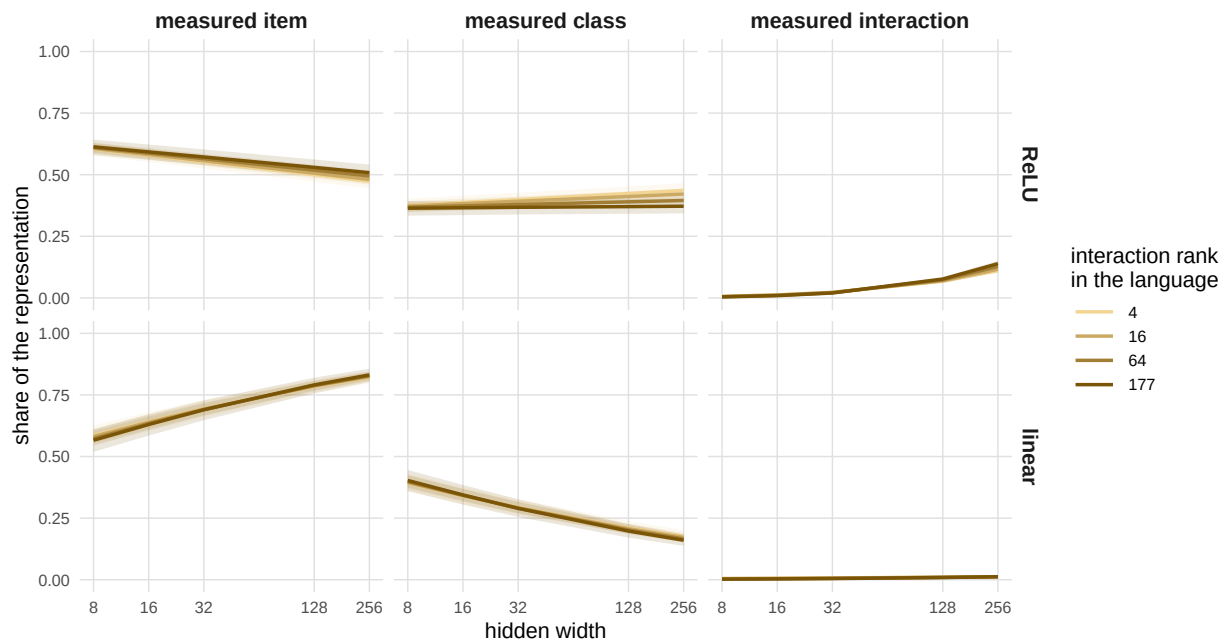


Figure 10: Fitted share of the representation taken by each component, against hidden width, with one line per rank of the interaction planted in the language and 95% intervals. Rows are the two activations, columns the three measured components. Effect sizes are held at their means. Were capacity the governing quantity, rank would separate the lines as strongly as width slopes them.

| family | model           | params | width | layers | training data            | tokens |
|--------|-----------------|--------|-------|--------|--------------------------|--------|
| BabyLM | opt-babylm-125m | 125M   | 768   | 12     | BabyLM (child-available) | ~3B    |
|        | opt-babylm-350m | 350M   | 1024  | 24     | BabyLM (child-available) | ~3B    |
|        | opt-babylm-1.3B | 1.3B   | 2048  | 24     | BabyLM (child-available) | ~3B    |
| Pythia | pythia-160m     | 160M   | 768   | 12     | the Pile                 | ~300B  |
|        | pythia-410m     | 410M   | 1024  | 24     | the Pile                 | ~300B  |
|        | pythia-1.4b     | 1.4B   | 2048  | 24     | the Pile                 | ~300B  |

Table 13: The six models, paired by width. Training-token counts are approximate; the OPT-BabyLM models train for 20 epochs over a 100M-word corpus, which is roughly 150M tokens, so their effective token exposure is about twenty times that. *Layers* counts the transformer blocks, and the measure is applied to each block’s output plus the embedding layer.

The six models are listed in Table 13. The two families are matched on width rather than on parameter count, which is what the measure is sensitive to: the chance level for overlap is the class subspace rank over the width (Appendix C), so a comparison across models of different width has to be read against different chance levels. Pairing at 768, 1024, and 2048 removes that from the family comparison.

They differ in two ways that matter elsewhere in the paper. The training corpora differ by more than two orders of magnitude in size, which is the manipulation of interest. And they position tokens differently: Pythia uses rotary embeddings, applied inside attention, while OPT learns absolute position embeddings added at the input. Appendix I uses that difference to rule out absolute position as the source of the interaction.

1004

1005

1006

1007

1008

1009

1010

1011

1012

**F   Stimulus Counts**

| class                               | item types | median tokens |
|-------------------------------------|------------|---------------|
| part of speech (noun / verb)        | 78         | 18 / 18       |
| grammatical role (subject / object) | 49         | 15 / 19       |

Table 14: Stimuli for the two linguistic categories, with the two classes of each in parentheses. *Item types* is how many word forms occur in both of them at least ten times; forms occurring in only one are excluded. *Median tokens* is how often a typical form occurs in each of its two classes, in the order they are listed.

## G Standardizing the Hidden States

Transformer hidden states carry a few dimensions whose variance exceeds the rest by two to three orders of magnitude (Timkey and van Schijndel, 2021), and we divide each dimension by its standard deviation before decomposing. The correction is there for the second measure rather than the first. A share is a ratio of squared lengths, and a diagonal rescaling scales the three components alike unless one of them loads disproportionately on the large dimensions, so the sizes are largely unaffected by it. An overlap is a projection onto a subspace estimated from the scatter, and the leading directions of that scatter are set by the largest dimensions, so without the correction every subspace tilts toward the same few axes and unrelated components appear to share directions. The random-subspace baseline does not absorb this, since a random subspace of matched rank is isotropic and inherits none of the tilt, leaving the ratio to chance inflated along with the overlap.

Table ef{tbl-standardize} reports both readings for part of speech, which was measured twice. The interaction’s share is unchanged to three decimals in every model. What moves is the overlaps, and they move in one family: item-class overlap is three to four times larger unstandardized in Pythia and identical to two decimals in OPT-BabyLM. That split is what a rogue-dimension account predicts, since massive activations are documented in the former and the latter is trained on a hundredth of the data; an account on which the correction removes class signal would have to explain why it removes none of it from three of the six models.

The one quantity we would not defend either way is Pythia’s class effect, whose share falls by 12 to 31% under the correction. Nothing in the paper turns on its absolute size, and the comparative claim that the interaction runs above it holds under both readings. Grammatical role was measured once, standardized, so the comparison here is for part of speech only.

We use a per-dimension rescaling rather than whitening or a principal-components truncation. Whitening needs the full covariance, which at 768 to 2048 dimensions and roughly 3,700 tokens per grid is near-singular, and making the total scatter isotropic would flatten the between-cell structure the decomposition exists to partition. A truncation discards directions irreversibly and needs a cutoff chosen by hand, and with two classes the class effect spans a single direction that such a cutoff could remove outright. Rotating to principal axes without rescaling changes nothing at all: the decomposition and every quantity we report from it are invariant under orthogonal transformation.

| model       | interaction |       | class effect |       | item-class overlap |        |
|-------------|-------------|-------|--------------|-------|--------------------|--------|
|             | std         | raw   | std          | raw   | std                | raw    |
| BabyLM 125M | 0.095       | 0.095 | 0.072        | 0.074 | 0.0052             | 0.0053 |
| BabyLM 350M | 0.122       | 0.122 | 0.066        | 0.068 | 0.0056             | 0.0061 |
| BabyLM 1.3B | 0.104       | 0.104 | 0.050        | 0.051 | 0.0035             | 0.0036 |
| Pythia 160M | 0.231       | 0.228 | 0.078        | 0.089 | 0.0058             | 0.0081 |
| Pythia 410M | 0.244       | 0.242 | 0.098        | 0.123 | 0.0067             | 0.0195 |
| Pythia 1.4B | 0.242       | 0.238 | 0.086        | 0.124 | 0.0055             | 0.0234 |

Table 15: Part of speech at the deepest layer, measured with and without the per-dimension rescaling. Sizes are shares of the item-by-class structure; overlaps are the raw fractions, not multiples of chance.

## H Deepest-Layer Measurements

Table 16 gives the deepest-layer values behind the curves of Figure 4, together with the null each overlap is read against.

| class            | model       | item-class overlap | interaction overlap | class effect         | interaction          |
|------------------|-------------|--------------------|---------------------|----------------------|----------------------|
| part of speech   | BabyLM 125M | 0.0052 (4.2×)      | 0.178 (3.5×)        | 0.072 [0.072, 0.072] | 0.096 [0.093, 0.098] |
|                  | BabyLM 350M | 0.0056 (5.9×)      | 0.166 (4.3×)        | 0.066 [0.066, 0.066] | 0.122 [0.119, 0.124] |
|                  | BabyLM 1.3B | 0.0035 (7.6×)      | 0.106 (4.8×)        | 0.050 [0.050, 0.051] | 0.104 [0.102, 0.106] |
|                  | Pythia 160M | 0.0058 (4.6×)      | 0.231 (4.2×)        | 0.078 [0.078, 0.078] | 0.232 [0.229, 0.235] |
|                  | Pythia 410M | 0.0067 (6.9×)      | 0.239 (6.1×)        | 0.098 [0.098, 0.099] | 0.245 [0.242, 0.248] |
|                  | Pythia 1.4B | 0.0055 (11.3×)     | 0.209 (9.7×)        | 0.086 [0.086, 0.086] | 0.243 [0.240, 0.246] |
| grammatical role | BabyLM 125M | 0.0054 (4.0×)      | 0.118 (3.5×)        | 0.050 [0.050, 0.050] | 0.045 [0.043, 0.048] |
|                  | BabyLM 350M | 0.0051 (5.5×)      | 0.102 (4.0×)        | 0.050 [0.049, 0.050] | 0.059 [0.055, 0.062] |
|                  | BabyLM 1.3B | 0.0033 (6.7×)      | 0.071 (4.7×)        | 0.039 [0.039, 0.039] | 0.065 [0.061, 0.069] |
|                  | Pythia 160M | 0.0049 (3.9×)      | 0.179 (5.3×)        | 0.078 [0.078, 0.079] | 0.092 [0.087, 0.095] |
|                  | Pythia 410M | 0.0047 (4.9×)      | 0.161 (6.5×)        | 0.074 [0.074, 0.075] | 0.087 [0.083, 0.091] |
|                  | Pythia 1.4B | 0.0062 (13.0×)     | 0.141 (11.2×)       | 0.078 [0.078, 0.079] | 0.094 [0.090, 0.098] |

Table 16: Deepest-layer measurements, one row per model. The two overlaps are the fraction of the item effect lying in the class subspace and the fraction of the interaction lying in the span of the two main effects, each followed in parentheses by how many times its own rotation null’s median it is. That null holds the effect fixed and rotates the target subspace at random 200 times. The class effect and the interaction are shares of the item-by-class structure, with each model’s re-split 95% interval in brackets.

### H.1 How much of a token the components account for

A share of the item-by-class structure answers how that structure divides. It does not answer how much of a word’s representation the structure is, and the two differ by about a factor of three, because most of what varies in a token’s hidden state is the context it appears in rather than anything about its word or its class.

Table ef{tbl-gain} puts the three components in one currency. Take a single token, *walk* used as a noun in some sentence, and ask how well its hidden state can be guessed. Guessing the grand mean is the baseline. Knowing that the word is *walk* removes the *word* column’s share of the squared error; knowing further that it is a noun removes the *class effect* column’s share; knowing what is specific to *walk-as-a-noun* removes the *interaction* column’s. The three components are orthogonal by construction, so each contributes its own term and the columns accumulate.

Two things follow. The class effect, the abstract category the representation is usually said to encode, accounts for 1% to 3% of a token; the interaction accounts for 2% to 8%, matching or exceeding it in 11 of the twelve grids. And nothing here exceeds 28% to 44% at the top end, because the remainder is within-cell context variation, which no item-by-class decomposition sets out to capture.

The last column states the interaction’s contribution the other way round, as a share of what the additive account leaves unexplained rather than of the whole. That is the quantity the additive account is judged on, and it runs 3% to 10%.

Read geometrically the same quantity looks smaller than it is. The displacement of a cell mean from where an additive account puts it is only about a fifth to a third of the radius of the token cloud around that mean, which invites the reading that the interaction lands inside the cloud and so cannot matter. In 768 to 2048 dimensions that reading does not hold: a cloud is a thin shell with essentially no tokens near its centre, so the observed cell mean fails the same test. Along the direction that actually separates the two classes of a word, the probes of Experiment 3 imply a standardized separation of roughly two to three.

| class            | model       | word | class effect | interaction | over additive |
|------------------|-------------|------|--------------|-------------|---------------|
| part of speech   | BabyLM 125M | 36.3 | 3.1          | 4.2         | 6.9           |
|                  | BabyLM 350M | 31.2 | 2.5          | 4.7         | 7.1           |
|                  | BabyLM 1.3B | 25.8 | 1.5          | 3.2         | 4.4           |
|                  | Pythia 160M | 22.4 | 2.5          | 7.5         | 10.0          |
|                  | Pythia 410M | 20.3 | 3.0          | 7.6         | 9.9           |
|                  | Pythia 1.4B | 18.8 | 2.4          | 6.8         | 8.7           |
| grammatical role | BabyLM 125M | 38.4 | 2.1          | 1.9         | 3.3           |
|                  | BabyLM 350M | 33.0 | 1.8          | 2.2         | 3.3           |
|                  | BabyLM 1.3B | 25.9 | 1.1          | 1.9         | 2.6           |
|                  | Pythia 160M | 28.0 | 2.6          | 3.1         | 4.4           |
|                  | Pythia 410M | 27.5 | 2.4          | 2.9         | 4.1           |
|                  | Pythia 1.4B | 24.9 | 2.4          | 2.8         | 3.9           |

Table 17: How much of a token’s hidden state each component accounts for, at the deepest layer, as the percentage reduction in squared error against guessing the grand mean. The three components are orthogonal, so their contributions add: *word* plus *class effect* is the additive account, and adding *interaction* gives the full decomposition. *Over additive* restates the interaction the other way round, as a percentage of the error the additive account leaves.

## I Positional Embeddings Are Not the Source

One architectural explanation of the interaction has to be ruled out before it can be read as a fact about lexical representations. The BabyLMs add a *learned absolute position embedding* at the input. The two classes of a linguistic category are not positionally interchangeable in a sentence, since subjects precede objects and nouns and verbs occupy different slots, so a representation that varies with position varies with the class. If that is what the decomposition is picking up, the interaction would be an artifact of where words sit rather than of how they are represented.

We re-extracted every OPT-BabyLM model with the positional embeddings zeroed out and measured it identically. The ablation was verified per model rather than assumed: with the embedding zeroed, the same token at two different offsets must give identical layer-0 states, and it does, in every model (drift 0.0 to machine precision). Pythia has no such module, so there is nothing to zero there.

We measure the class effect’s share of class-relevant structure which is the class effect divided by the sum of the class effect and the interaction, which we call the *portable fraction*. If positions were manufacturing the interaction, removing them should leave a larger share of what distinguishes the two classes in a direction shared across words. Absolute sizes cannot be compared here: zeroing the embeddings in a model that was trained with them is a heavy intervention, and it shrinks every component at once, the class effect included.

| model           | positions | class | interaction | portable fraction |
|-----------------|-----------|-------|-------------|-------------------|
| opt-babylm-125m | zeroed    | 0.009 | 0.026       | 0.258             |
| opt-babylm-125m | intact    | 0.072 | 0.095       | 0.430             |
| opt-babylm-350m | zeroed    | 0.011 | 0.043       | 0.205             |
| opt-babylm-350m | intact    | 0.066 | 0.122       | 0.351             |
| opt-babylm-1.3B | zeroed    | 0.007 | 0.032       | 0.184             |
| opt-babylm-1.3B | intact    | 0.050 | 0.104       | 0.327             |

Table 18: Trained OPT-BabyLM at the deepest layer, with absolute position embeddings intact and zeroed. Sizes are shares of the item-by-class structure and the portable fraction is the class effect divided by the sum of the class effect and the interaction.

Removing the position embeddings makes these representations *less* separable, not more. The portable fraction falls in every model, from 0.430 to 0.258, from 0.351 to 0.205, and from 0.327 to 0.184 (Table 18). Whatever the absolute position embedding contributes, it is contributing to the part of class structure that *is* carried by a shared direction. Take it away and what remains is more word-specific, not less. The interaction is not an artifact of position.